Household Vehicle Purchase Timing: Findings from the Literature

Evidence, Model Assumptions, and Implications for the Auto-Loan Research Exercise

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Abstract

This review assesses what the literature establishes about household vehicle-purchase timing and what can inform the purchase decision preceding auto-loan demand. Studies of durable adjustment and vehicle transactions link purchase timing to existing ownership, transaction costs, and changing household circumstances. Consumer-credit studies show that liquidity and offered terms affect transactions, although evidence from applicants and borrowers does not directly identify purchase probabilities for all households. Dynamic choice models formalize the value of waiting; their interpretation depends on assumptions about state transitions, unobserved heterogeneity, and initial conditions. Identification results further show that observed choices need not recover individual utility components or responses to every policy change. These findings support comparing state-dependent purchase models, with financial feasibility made explicit, against simpler timing models. They do not establish a unique Need–Ability–Choice specification or a forecasting advantage for structural estimation. For the JPMorgan exercise, the unresolved requirements are the purchase endpoint, observable household histories, decision frequency, and the intended use of the forecast. Central equations and compatibility conditions are presented in the main text; detailed derivations and source checks are retained in the appendices.

1 Research Question and Principal Findings

The immediate research question is which information and economic mechanisms can explain or predict a household's purchase of a vehicle within a future six- or twelve-month horizon. The broader JPMorgan exercise concerns potential auto-loan demand among existing bank customers. A vehicle purchase, borrowing to fund it, and originating a loan at a particular bank are distinct outcomes. This review concerns the purchase event, including additions and replacements bought with cash or credit. New leases are excluded from the working purchase endpoint. Lender selection, repayment risk, and a separate model of household risk attitudes remain outside the current task; liquidity and offered financing terms enter as potential determinants of whether and when a purchase is feasible.

Three conclusions organize the review. First, vehicle-purchase timing depends on the household's starting ownership position and on the cost of continuing with it. Durable-adjustment and transaction studies support this conclusion through different outcomes and data, without establishing a universal life-event trigger (Eberly, 1994; Attanasio, 2000; Clark et al., 2016; Yamamoto et al., 1999). Second, liquid resources and financing terms affect transactions, but the strongest evidence often concerns selected applicants or borrowers (Adams et al., 2009; Attanasio et al., 2008; Argyle et al., 2021). Third, a model's ability to fit purchase histories does not establish that its latent mechanisms, utility components, or policy responses are identified (Magnac and Thesmar, 2002; Heckman, 1981; Arcidiacono and Miller, 2020). Each conclusion has a direct consequence for the proposed application:

record the initial fleet, distinguish financial resources from observed loan outcomes, and match the model's claims to what the data and assumptions can identify.

The studies were selected for their relevance to purchase timing, financial feasibility, persistent or changing household differences, and the identification of dynamic choices. This is a selective review, not an exhaustive search establishing that no alternative approach exists. Need, Ability, and Choice provide an organizing interpretation of adjustment pressure, feasible transactions, and decisions among alternatives. The reviewed evidence does not establish three separately identifiable latent equations, a fixed sequence of decisions, or independence among the mechanisms.

The main text compares empirical findings, presents the equations that determine the models' relevance, and states the conditions under which components can be combined. Appendix A retains the detailed paper analyses in the authors' notation, including source locations, reconstructed steps, and source-check observations. Appendix B records additional probability representations used in the earlier formulation. Appendix C separates incomplete readings and candidate references from the completed review. The companion proposal remains a provisional formulation. The current specification task is to compare progressively richer utility models; neither the decision interval nor an estimator is settled by this review.

2 Empirical Evidence on Vehicle-Purchase Timing

2.1 Ownership and household changes affect different transaction margins

The durable-goods literature supports adjustment costs as an explanation for intermittent automobile purchases. Eberly (1994) finds evidence consistent with an (S, s) adjustment rule for a substantial subset of households, while Attanasio (2000) studies estimation and aggregation when those adjustment rules differ across households. These studies connect periods of inaction to the distance between current and preferred durable holdings and to the cost of changing them. Their contribution is a mechanism for delayed adjustment; they do not identify a directly observed measure of a household's need to purchase.

The transportation evidence shows why starting ownership must accompany lifecycle variables. Yamamoto (2008) relates vehicle transactions and ownership changes to changes in adults, employment, and residence, using French and Japanese data with different outcomes. The improvement in fit from life-course covariates is relatively modest in that study. Clark et al. (2016) distinguish zero-to-one, one-to-two, and downward transitions in a UK household panel and finds that associations with household and spatial characteristics vary with initial ownership. These results support state-specific responses to household events. They do not justify assigning the same purchase response to an employment change in a car-free household and in a household already owning two vehicles.

Timing also matters for interpretation. Oakil et al. (2014) find that ownership changes can precede anticipated childbirth or employment transitions. Gu et al. (2021) permit heterogeneous responses through latent classes in a competing-risks model. Consequently, an event observed near a purchase can be part of an anticipated adjustment, and an average association may conceal different household responses. A purchase forecast must use information available at its origin; retrospectively recorded events do not automatically supply usable predictors.

Recent studies cover narrower endpoints that should remain distinct. Oshanreh et al. (2025) examine purchase timing using the Panel Study of Income Dynamics, where interval-observed purchases constrain temporal precision. Hossain and Fatmi (2025) study first purchases by car-free households and relate acquisition timing to vehicle alternatives through a logsum. First purchase, replacement, and an increase in the number of vehicles can share explanatory variables while

requiring different risk sets and state transitions. For the JPMorgan exercise, the evidence supports recording existing holdings and distinguishing transaction types before deciding how much behavioral detail a model needs.

2.2 Liquidity affects purchases, but the evidence is sample-specific

Adams et al. (2009) provide direct evidence that up-front liquidity matters within a subprime used-car channel. The study observes more than 50,000 applications at one dealer-lender, with approximately 34 percent resulting in a purchase. In Table 2, column 2, a $100 increase in the required minimum down payment reduces the estimated purchase probability by 0.0299, with a standard error of 0.0006. The corresponding financed-price effect is much smaller and imprecisely estimated. Applications and sales also increase during tax-refund season. These results distinguish cash required at purchase from the price financed over time.

The finding is conditional on applying at that dealer-lender. Changes in minimum down payments must be exogenous to demand after the study's controls; negotiated prices require an exclusion restriction, and nonpurchaser prices are imputed. The reported price-equivalent and discount-rate calculations depend on an imprecise price coefficient. The down-payment estimate is therefore a clearer empirical finding to carry into the synthesis than a precise claim about the implied discount rate. The study supports a liquidity mechanism; its coefficients cannot be inserted into a general household purchase model without a separate transfer argument. Appendix A.4.1 records the equations and identification details.

Attanasio et al. (2008) find substantial responsiveness of car-loan demand to maturity and weaker responsiveness to interest rates except among high-income households, interpreting the pattern as evidence consistent with borrowing constraints. Argyle et al. (2021) show that shorter offered maturities reduce used-car expenditure substantially through substitution toward lower-quality vehicles, with additional adjustment through negotiated prices. The two studies establish that loan terms can change both borrowing demand and the transaction selected. Neither outcome is identical to the probability of any vehicle purchase by a household over the next six months.

The form of income matters as well. Jiang et al. (2021) associate lump-sum bonuses with greater demand for auto loans and report delinquency responses that differ from those following increases in base salary. For the present review, the relevant finding is the distinction between temporary funds and recurring income; subsequent repayment behavior belongs to a separate target. Together, these studies support treating accessible cash, payment obligations, trade-in equity, and offered terms as different financial variables. They do not establish that annual income or a credit score alone measures the feasible transaction set.

The generalization problem is substantive. Originated loans omit households that do not apply, rejected applicants, cash purchases, and purchases financed elsewhere. A model trained on those records may describe loan origination in the observed channel while missing the broader purchase event. Whether the eventual bank data identify the latter must be established from their coverage, not inferred from the availability of detailed loan records.

2.3 Product choice and purchase timing answer different questions

The differentiated-product literature explains substitution among vehicles. McFadden (1974) supplies a random-utility foundation for discrete choice, and Berkovec and Rust (1985) apply nested logit to the holdings of one-car households. Berry (1994) and Berry et al. (1995) address aggregate shares, heterogeneous preferences, and price endogeneity. These contributions inform the definition and valuation of alternatives. Aggregate market shares, however, do not directly recover a named

household's current liquidity or prospective purchase date.

Dynamic automobile studies add the option to wait. Adda and Cooper (2000a) show how scrappage incentives can bring purchases forward and alter later activity through the distribution of vehicle ages. Schiraldi (2011) uses holding and purchase shares to estimate a dynamic replacement model with transaction costs. Its distinction between retaining a car and purchasing another allows the same vehicle attributes to enter differently for owners and prospective buyers. The resulting evidence supports including the ownership state and continuation alternative, while the aggregate identification strategy does not transfer automatically to customer-level records.

The combined implication is that an attractive alternative and the timing of acquisition can be related without households following a fixed sequence of independent decisions. Credit terms can change the vehicle chosen; anticipated future terms can change when the household buys. A reduced-form timing model can represent predictive associations among these variables. A structural account additionally needs a defensible specification of what the household knows, what it can choose, and how its choices change future opportunities.

2.4 Transaction histories determine the probability being estimated

Vehicle-transaction studies provide a more direct empirical precedent for timing than static ownership models. de Jong (1996) jointly models holding duration, vehicle type, and use. Yamamoto et al. (1999) distinguish replacement, disposal, and additional acquisition as competing events; Mohammadian and Rashidi (2007) compare independent and competing-hazard formulations using changing household characteristics and a vehicle-alternative logsum. These studies connect transaction timing to household states and available alternatives without estimating separate latent Need, Ability, and Choice equations.

Later work addresses discrete observation intervals and heterogeneity (Rashidi and Mohammadian, 2011a), interdependence with residential and employment changes (Rashidi and Mohammadian, 2011b; Rashidi et al., 2011), and dependence among transaction disturbances through a copula (Hossein Rashidi and Mohammadian, 2016). Ghasri et al. (2018) compare duration and discrete-choice approaches to time-dependent ownership outcomes. These are alternative treatments of timing, sampling, and dependence. Their existence does not establish a uniformly preferred architecture or identify three latent economic mechanisms.

The distinction between first-event and any-purchase probabilities is consequential for the application. The following identities use the review's notation and a monthly representation solely to make the endpoints explicit; they do not select the model's eventual decision interval. At forecast origin t_0, let $\widetilde{T}_h$ be the month of the first transaction of any type and J_h its mutually exclusive type $k \in \mathcal{K}$. Conditional on origin information $\mathcal{I}_{h,t_0}$, define

$$\lambda^{\mathrm{tr}}_{hk}(r; t_0) = \mathbb{P}\Big(\widetilde{T}_h = t_0 + r, J_h = k \mid \widetilde{T}_h \geq t_0 + r, \mathcal{I}_{h,t_0}\Big). \tag{1}$$

Then survival without a transaction and the incidence of a first transaction of type k are

$$S^{\mathrm{tr}}_h(r; t_0) = \prod_{u=1}^{r} \left\{ 1 - \sum_{k \in \mathcal{K}} \lambda^{\mathrm{tr}}_{hk}(u; t_0) \right\}, \qquad S^{\mathrm{tr}}_h(0; t_0) = 1, \tag{2}$$

$$F^{\mathrm{tr}}_{hk}(t_0, H) = \sum_{r=1}^{H} S^{\mathrm{tr}}_h(r-1; t_0) \lambda^{\mathrm{tr}}_{hk}(r; t_0). \tag{3}$$

For purchase types $\mathcal{P} \subseteq \mathcal{K}$, summing F^{tr}_{hk} over $k \in \mathcal{P}$ gives the probability that the *first transaction is a purchase*. A disposal followed by a purchase contributes to the probability of *any purchase*

within the horizon but not to that first-event quantity. Competing risks therefore affect the endpoint as well as the estimator (Kiefer, 1988; Andersen et al., 2012). Appendix B records the explicit endpoint identity and the earlier framework representations.

The hazards above integrate later histories within the risk set using information at t_0. If hazards instead condition on evolving covariates, those histories must be integrated to obtain a prospective forecast. Annual or biennial observations may interval-censor the transaction date; pre-existing ownership spells and incomplete follow-up create additional entry and censoring issues. Multiple transactions require an event process consistent with the chosen endpoint. A more detailed model cannot recover timing that the observations do not identify.

2.5 Customer histories support prediction without establishing intervention effects

Adjacent marketing studies examine purchase timing with incomplete histories and persistent heterogeneity (Jain and Vilcassim, 1991; Vakratsas and Bass, 2002). Rossi et al. (1996) demonstrate the value of purchase histories for targeting. These studies motivate using customer histories, but they are not direct evidence about vehicle replacement or the incremental effect of a bank offer.

The banking study by Li et al. (2011) is closer in data setting. In the authors' November 2010 manuscript, a model of latent financial states and monthly product openings achieves a validation purchase hit rate of 0.268, compared with 0.101–0.246 for the reported alternatives. Its simulated profit improvement is evaluated under the fitted model, not through a field intervention. A bank-product opening is also a different event from a vehicle purchase. The useful contribution is a concrete treatment of changing latent states and observed banking histories; identifying those states and transferring their interpretation remain separate questions, assessed in Section 3.4.

3 What the Selected Models Establish

3.1 Replacement models connect current costs to the value of waiting

Rust (1987) establishes an estimable model of repeated replacement by making current operating cost and expected future cost depend on the existing asset. In his notation, x is mileage, $i = 0$ means keep, $i = 1$ means replace, and RC is replacement cost. With additive, independent type-I extreme-value shocks and the paper's conditional-independence restriction, equations (4.11)–(4.14) give

$$P(i \mid x, \theta) = \frac{\exp\{u(x, i, \theta_1) + \beta EV_\theta(x, i)\}}{\sum_j \exp\{u(x, j, \theta_1) + \beta EV_\theta(x, j)\}}. \tag{4}$$

The continuation term $EV_\theta(x, i)$ depends on the transition induced by the action. It is this dependence that distinguishes the decision from a static comparison of present costs. The result requires the observed state to carry the relevant history under the maintained shock and transition assumptions. The empirical application is a maintenance manager's decisions for 104 buses, not household purchases. Rust's tests also reject conditional independence for some bus groups, so the computationally convenient restriction is not uniformly supported even in that application. Appendix A.1.1 gives the original utilities, transition law, and likelihood.

Schiraldi (2011) extends the replacement mechanism to new and used automobiles with resale and transaction costs. In the author's notation, the purchase alternatives are summarized by a dynamic inclusive value δ_{it}; with denominator D_{it} including purchase, keep, and exit options,

summing equations (11)–(12) yields

$$\sum_{j \in J_t} d_{it}^{kj} = \frac{\exp(\delta_{it})}{D_{it}}. \tag{5}$$

Here $D_{it} = \exp(\delta_{it}) + \exp(W_{it}^0) + I(k \neq 0)\exp(W_{it}^k)$, with W_{it}^0 and W_{it}^k denoting the exit and keep values in the appendix's shorthand. The summation is the review's reconstruction: it shows how the attractiveness of a vehicle menu can enter the timing margin without treating purchase timing and product choice as independent. The model uses annual aggregate data from Isernia, Italy, with a fixed discount factor and a one-car restriction. Its state reduction imposes Markov beliefs about the inclusive value and the held car's value. The share inversions recover transaction costs under these restrictions; their identification argument and aggregate data cannot simply be substituted for household histories. Appendix A.1.2 records the original definitions and the limits of the inversion argument.

3.2 A budget constraint changes the replacement problem

Gavazza and Lanteri (2021) contribute a mechanism missing from the preceding replacement models: liquid wealth, income risk, and a borrowing limit jointly determine consumption, saving, and the car held. In their notation, b is current bond wealth, w is income, n is the current car, $\widetilde{n}$ is the chosen car, p_b is the bond price, and τ is a tax. Their equations (3), (5), and (6) combine the Bellman problem with

$$V(b, w, n; \theta) = \max_{c, b', \widetilde{n}} \left\{ u(c, d(\widetilde{n}, \theta)) + \beta E[V(b', w', n'; \theta) \mid \widetilde{n}, w] \right\}, \tag{6}$$

$$c + p_{\widetilde{n}} + p_b b' + \tau = w + p_n - \lambda(p_n) I(\widetilde{n} \neq n) + b, \qquad b' \geq \phi. \tag{7}$$

Here λ is the selling cost and θ a permanent type. The budget makes the effect of used-car prices depend on both the price of the next vehicle and the value of the current one. In the calibrated equilibrium exercise, tighter credit reduces purchases partly because constrained owners delay scrapping, lowering used-car values and changing upgrade costs for other households.

This is a calibrated annual equilibrium model with joint discrete car choice and continuous saving, rather than an estimated logit choice model. Its household policy is deterministic conditional on the full state and type; the calibration does not specify a statistical household-choice likelihood. Adding choice shocks would be a modeling extension requiring a new solution and estimation argument. The common borrowing limit also does not represent an amortizing auto-loan contract or lender pricing. The contribution to the present task is the explicit connection between feasible purchases, saving, and trade-in equity. The calibrated magnitudes and preference curvature do not transfer automatically to the intended population. Appendix A.1.3 retains the calibration, policy, and equilibrium details.

3.3 Choice probabilities do not identify every structural object

Magnac and Thesmar (2002) show why observed choices and state transitions need not identify utilities, the discount factor, and the shock distribution jointly. Hotz and Miller (1993) supply an inversion from conditional choice probabilities (CCPs) to value differences, conditional on the shock specification, and use it to reduce the computation required for estimation. These are complementary results: recovering value differences does not remove the need to fix or identify the remaining primitives. Hotz–Miller's terminating-action simplification also cannot be assumed for a vehicle

purchase that leaves persistent financial states unchanged or only partly reset. Appendices A.2.1 and A.2.2 give the original restrictions and equations.

The normalization issue is explicit in Arcidiacono and Miller (2020). In the stationary case, their equation (9) gives observationally equivalent payoff vectors

$$u_j^* = u_j + [I - \beta F_j][I - \beta F_l]^{-1}(c - u_l), \tag{8}$$

where F_j is the state-transition matrix under action j, l is a reference action, and c is the payoff assigned to it. The equivalence holds for the maintained discount factor, transitions, and shock distribution. In a dynamic model, changing the reference payoff changes how future payoffs enter the recovered utilities; normalized coefficients therefore need not represent the same economic cost components.

Counterfactual identification depends on what changes. For nonstationary short panels, Arcidiacono–Miller's Theorem 5 identifies responses to known temporary payoff innovations that end within the observed horizon, under the maintained primitives. A known payoff for one action is generally insufficient for transition innovations in short panels; additional conditions, including single-action finite dependence in their treatment, matter. In the stationary entry-exit setting of Aguirregabiria and Suzuki (2014), some transition changes outside the scrap-value component are admissible under Proposition 3, and Proposition 7 permits further transition counterfactuals under added exclusions. A single blanket verdict about all payoff or transition changes would erase these distinctions.

For the purchase problem, an offer that changes only a known current payoff is different from one that also changes future liquidity, feasible borrowing, or the transition of other payoff-relevant states. The review establishes that this distinction must be specified before interpreting a policy response. It does not establish that a particular bank offer satisfies the identifying conditions. Appendices A.2.3 and A.2.4 distinguish the source results from the review's reconstructions.

3.4 Persistent types and changing latent states require different evidence

Kasahara and Shimotsu (2009) supply conditions for identifying finite mixtures of dynamic discrete choices. In the working-paper version reviewed here, the sequence probability is

$$\mathbb{P}(\{a_t, x_t\}_{t=1}^T) = \sum_{m=1}^{M} \pi^m p^{*m}(x_1, a_1) \prod_{t=2}^{T} f^m(x_t \mid x_{t-1}, a_{t-1}) P^m(a_t \mid x_t, a_{t-1}). \tag{9}$$

The permanent type m affects the initial distribution, transitions, or choices under the relevant version of the model. Identification comes from restrictions on these objects and from rank conditions; the mixture notation alone provides no identification result. Their baseline uses full-support transitions common to types, which a deterministic vehicle-age transition violates. Extensions have their own support and panel-length requirements.

Higgins and Jochmans (2023) identify mixtures of time-homogeneous Markov chains from four observations under rank, distinguishability, and connectivity conditions without requiring every transition to be positive. That removes one obstacle to representing deterministic state components, but does not establish that a household dataset has sufficient variation. Rare purchases and sparse state cells raise a finite-sample concern that has not been evaluated for this application. The result also concerns permanent types, not an unrestricted sequence of changing latent household states.

Aguirregabiria et al. (2021) take a different route by conditioning out unrestricted fixed effects. Its advantage is reduced reliance on a parametric heterogeneity distribution. The cost is limited identification of the desired payoff objects: in the forward-looking binary model with unrestricted

duration dependence, no structural parameter is identified by that conditional likelihood; under the paper's flattening restriction, a duration increment becomes identifiable. Replacement-cost levels and the type distribution are not thereby recovered. The review's limited enumeration of informative histories is an illustration, not a general theorem about every household requiring two purchases. Appendices A.2.5–A.2.7 give the separate results.

The latent financial state in Li et al. (2011) changes over time and is measured through product openings. The reviewed manuscript combines state-specific multivariate-probit opening probabilities with semi-Markov state durations and persistent household heterogeneity. It treats households as myopic conditional on the state, while the bank's solicitation policy is dynamic. Combining that measurement structure with forward-looking household choice would require specifying the household's information and a consistent latent-state likelihood; it is not an immediate substitution into Rust's observed-state model. Duration dependence also requires elapsed time in the state or an equivalent representation. Appendix A.3.1 records the manuscript version and its reporting limitations.

3.5 Panel entry determines the appropriate heterogeneity distribution

Heckman (1981) shows that treating the first observed state as independent of persistent heterogeneity generally changes the likelihood and biases dynamic estimates. Keane and Wolpin (1997) handle initial schooling by allowing type shares to depend on that state. In contrast, Card and Hyslop (2005) study a sample with common initial welfare states. These are different ways to condition the analysis on the population actually observed, not interchangeable solutions for any panel.

The review's reconstruction makes their connection explicit. For a finite type k, initial state s_0, and subsequent history likelihood L_{ik}, the conditional contribution is

$$L_i(\text{history}_i \mid s_{i0}) = \sum_k \pi_{k|s_{i0}} L_{ik}. \tag{10}$$

Keane–Wolpin permits initial-state-specific shares; a common-state sample has one conditioning cell. This identity does not prove identification of the mixture or justify ignoring other determinants of sample entry. Restricting a sample can be valid for the resulting population, but it changes the target. For bank records, entry through an auto-loan application or origination could itself depend on the purchase decision. The required conditioning and selection assumptions must therefore be determined from the data-generating process. Appendices A.5.1 and A.5.2 preserve the source likelihoods and the review's derivation.

4 Compatibility of the Reviewed Components

The papers support complementary mechanisms, but they differ in populations, time scales, information, and identification targets. Table 1 records the conditions that govern their possible combination. The verdicts assess the published structures; they do not report performance of an implemented household model.

Table 1: Compatibility and limits of transfer to household purchase timing.

Components	Assessment	Conditions and limits
Rust and Schiraldi	Related dynamic structure	Both use EV1 shocks and conditional independence. Schiraldi adds a vehicle menu, resale, and a restricted inclusive-value state; its aggregate-share identification does not transfer to household panels.
Replacement and liquidity (Gavazza and Lanteri)	Requires a new specification	Joint car choice and continuous saving require a consistent budget, preference curvature, and solution. Adding EV1 choice shocks is an extension. Period length must be selected before transitions and payoffs are adapted.
Forward-looking choice and Li et al.'s latent state	Requires a joint information and observation model	A persistent omitted state can invalidate the observed-state conditional-independence restriction. The household's knowledge of the state, semi-Markov duration, and the measurement likelihood must be specified together.
Adams et al. and a structural household model	Evidence on a mechanism	The static probit concerns subprime applicants at one dealer-lender. It supports a liquidity mechanism; it does not validate an unestimated household model or supply transferable coefficients.
Replacement and Kasahara–Shimotsu types	Support restrictions must be checked	Deterministic age violates the reviewed baseline's full support. Limited-support and type-specific-transition extensions impose separate conditions; applicability has not been established for the proposed state.
Replacement and Higgins–Jochmans types	Possible under rank and connectivity conditions	Zeros in transitions are allowed. Type distinguishability and initial-state information remain necessary. Precision with rare purchases and many states is an unresolved empirical issue.
Type mixture and Keane–Wolpin initial shares	Compatible conditional on entry assumptions	Saturated shares conditional on the first state reproduce a conditional mixture likelihood. A large or continuous initial state requires additional restrictions; endogenous entry may require a selection model.
Purchase panel and Card–Hyslop common-state sample	Valid only for a justified restricted population	A common initial state can define the sampled population. It does not eliminate selection issues or recover results for all bank customers; the conditioning event and full Markov state must be appropriate.
Replacement and Aguirregabiria–Gu–Luo fixed effects	Identifies a limited subset of payoffs	Under forward-looking duration dependence, the relevant identification uses flattening restrictions. Cost levels and type distributions are not recovered. The number of informative household histories requires a separate assessment.

Components	Assessment	Conditions and limits
Stationary normalization and counterfactuals	Depends on the changed primitives	Aguirregabiria–Suzuki permits known additive payoff shifts and specified transition changes outside the scrap-value component under Proposition 3. Additional exclusions expand the class under Proposition 7. General transition or discount-factor changes are not secured by normalization.
Nonstationary short panels and counterfactuals	Temporary payoff results have a bounded domain	Arcidiacono–Miller's Theorem 5 concerns known payoff changes ending within the observed horizon. For transition changes, knowing one action's payoff is generally insufficient without additional conditions such as single-action finite dependence.

Period conversion is a substantive modeling issue. Under constant exponential discounting, an annual factor of 0.9 corresponds to approximately 0.991 per month, but this does not convert an annual transition law, action menu, or payoff into a monthly model. Likewise, the commonly used rough probit-to-logit coefficient scaling of about 1.6–1.8 is an approximation, not a transfer rule between different samples or economic specifications. Comparisons must align the outcome, time interval, shock scale, and information set before interpreting parameter differences.

5 Implications for the JPMorgan Research Question

The reviewed literature supports a household state that records existing ownership and financial feasibility, with purchase compared against continuation. It also supports simpler event-history models as a serious forecasting comparison. What remains unresolved is whether the intended data can identify the additional behavioral objects of a structural model and whether those objects are needed for the decision the forecast will inform.

Define the target before choosing the model. The working endpoint includes vehicle additions and replacements, whether paid for in cash or financed. A loan-originations label would answer a narrower question. Household linking, transaction codes, and coverage of purchases made outside the bank must be assessed explicitly. The first-transaction probability in Section 2.4 is also narrower than the any-purchase target whenever a household can dispose of a vehicle and later buy within the horizon. These differences determine the population and risk set rather than constituting minor coding choices.

Use the utility sequence to expose assumptions. A static comparison can establish the state, feasible actions, and current payoff. A dynamic extension is justified when the action changes future attainable utility; a financing extension is justified when observed resources or terms constrain the transaction. Each extension should state what it adds and what information it requires. The literature supplies candidates for these components, but it does not select the final utility function or establish that a keep-or-replace special case covers first purchases and additions.

Choose the period jointly with the observations and forecast horizon. Monthly, quarterly, and semiannual decision periods remain alternatives. Coarsening the period reduces the number of nonpurchase intervals without creating new purchases, and it can conceal the order of household

events. The decision step, observation frequency, precision of the purchase date, and six- or twelve-month forecast horizon must be distinguished. Data support for those quantities should determine the comparison; the monthly frequency of some source papers does not settle it.

Treat latent mechanisms as hypotheses requiring evidence. Fleet condition, lifecycle changes, balances, payment obligations, offers, and vehicle equity may proxy for adjustment pressure or financial feasibility. Observed nonpurchase alone cannot distinguish a satisfactory current fleet, a binding constraint, or a high value of waiting. Repeated observations help only under explicit measurement, transition, and selection assumptions. Need–Ability–Choice is useful for organizing these questions, but a separate latent equation for each mechanism requires evidence beyond the conceptual decomposition.

Evaluate forecasting and intervention claims separately. For prediction under existing conditions, the relevant comparison is whether added structure improves prospective purchase probabilities relative to a simpler timing model using the same information. Evaluation should respect forecast dates, censoring, event definitions, and calibration across horizons. This is a proposed criterion, not an established result of the reviewed studies. If the intended decision is to change an offer, a further argument must identify the offer's effect and the utility or transition objects it changes. Purchase propensity, responsiveness to an offer, lender selection, and resulting profit are different outcomes.

The next evidence needed is therefore concrete: observable household and vehicle histories, purchase-label coverage, financial information available before the forecast origin, and a specified use for the probability. Once these are documented, the utility alternatives can be assessed against the compatibility conditions in Section 4. The review supports this sequence of research decisions; it does not establish a new empirical result, a uniquely identified three-layer model, or a forecasting advantage in the JPMorgan population.

A Technical Analyses of the Selected Papers

This appendix preserves the detailed equations, derivations, results, and source checks supporting the main text. The entries retain the authors' notation and source equation numbers; reconstructed arguments and invented numerical illustrations are identified separately. Source-check observations refer to the particular versions recorded in each entry and should not be read as claims about every version of the paper. The C1–C5 group labels identify replacement, identification, latent states, liquidity, and initial conditions. Incomplete readings are collected in Appendix C; the cross-paper assessment is in Section 4.

A.1 C1. Dynamic choice of when to buy

A.1.1 Rust (1987): bus engine replacement

Why it is included: it is the core of a purchase-timing model, a monthly keep-or-replace decision whose choice probability is a logit in current utility plus discounted future value.

Setup. One decision maker (the maintenance manager) and 104 buses, monthly, December 1974 to May 1985 (p. 1001). *Given state:* accumulated mileage since the last replacement, x_t, observed by both agent and researcher; cost shocks $\epsilon_t = (\epsilon_t(0), \epsilon_t(1))$, observed only by the agent. *Action:* $i_t \in C(x_t) = \{0, 1\}$, keep or replace. *Objective:* minimize expected discounted cost.

Equations and derivation.

1. **Flow utility** (5.1), p. 1015:

$$u(x_t, i, \theta_1) + \epsilon_t(i) = \begin{cases} -RC - c(0, \theta_1) + \epsilon_t(1) & i = 1, \\ -c(x_t, \theta_1) + \epsilon_t(0) & i = 0. \end{cases}$$

   Assumptions used: shocks enter additively; operating cost c depends only on mileage; normalization $c(0, \theta_1) = 0$, because adding a constant to utility does not change choices (p. 1015).

2. **Transition** (5.2), p. 1015: $p(x_{t+1} \mid x_t, i, \theta_3) = g(x_{t+1} - x_t, \theta_3)$ if $i = 0$ and $g(x_{t+1} - 0, \theta_3)$ if $i = 1$. Replacement resets mileage, so the process "regenerates." *Assumptions used:* first-order Markov mileage; increments independent of the shocks; discretized into 90 cells of 5,000 miles.

3. **Dynamic program** (4.1)–(4.4), pp. 1010: $V_\theta(x_t, \epsilon_t) = \max_i [u(x_t, i, \theta_1) + \epsilon_t(i) + \beta EV_\theta(x_t, \epsilon_t, i)]$, with a stationary optimal policy $i_t = f(x_t, \epsilon_t, \theta)$ (4.3). *Assumptions used:* infinite horizon, stationarity, discounting with $\beta < 1$, and regularity conditions in Rust's companion paper.

4. **Conditional independence (CI)** (4.7), p. 1011: $p(x_{t+1}, \epsilon_{t+1} \mid x_t, \epsilon_t, i) = q(\epsilon_{t+1} \mid x_{t+1})\, p(x_{t+1} \mid x_t, i)$. *Consequence:* EV_θ depends on (x, i) only, not on ϵ. *Assumption used:* next period's shock depends on the past only through x_{t+1}, and x_{t+1} does not depend on ϵ_t. Rust calls this "a strong assumption" (p. 1011). This is the step that makes the model computable.

5. **Choice probability** (Theorem 1, (4.8)–(4.10), p. 1011): $P(i \mid x, \theta)$ is the derivative of the expected maximum ("social surplus") G with respect to $u(x, i)$, and EV_θ is the unique fixed point of a contraction. *Assumption used:* CI plus a known shock density q.

6. **Extreme-value case** (4.11)–(4.14), p. 1012. With i.i.d. type-I extreme-value shocks, $G = \log \sum_j \exp[u(x, j) + \beta EV(x, j)]$ and

$$P(i \mid x, \theta) = \frac{\exp\{u(x, i, \theta_1) + \beta EV_\theta(x, i)\}}{\sum_j \exp\{u(x, j, \theta_1) + \beta EV_\theta(x, j)\}},$$

$$EV_\theta(x,i) = \int_y \log\Big\{ \sum_{j \in C(y)} \exp[u(y,j,\theta_1) + \beta EV_\theta(y,j)] \Big\} p(dy \mid x, i, \theta_3).$$

Assumptions used: EV1 distribution, independent across alternatives and time; mean and variance normalized (pp. 1015–1016). *Check:* if $p(\cdot \mid x, i)$ does not depend on i, then $EV(x,0) = EV(x,1)$ cancels and the static logit results (p. 1012). *Note:* as printed, the location in (4.11) is $+\gamma$, which conflicts with the mean-zero normalization on p. 1015; it changes only the level of EV, not P.

7. **Likelihood** (Theorem 2, (4.15), p. 1013):

$$\ell(x_1, \ldots, x_T, i_1, \ldots, i_T \mid x_0, i_0, \theta) = \prod_{t=1}^{T} P(i_t \mid x_t, \theta)\, p(x_t \mid x_{t-1}, i_{t-1}, \theta_3).$$

Past choices affect current choices only through x_t ("structural state dependence," p. 1013). *Assumptions used:* CI; buses independent of one another (fn. 11); the initial state (x_0, i_0) is taken as given.

8. **Estimation** (pp. 1016–1019): stage 1 estimates θ_3 from the transition factor alone; stage 2 estimates (θ_1, RC) from the choice factor, solving the fixed point (4.14) at every trial parameter (nested fixed point); stage 3 runs full maximum likelihood for efficient standard errors.

9. **Identification** (p. 1023, fn. 12; Table VIII, p. 1020). The cost level and the shock location and scale are normalized. β is collinear with RC (both delay replacement), so it is fixed at .9999 or 0. With a nonparametric cost function the log-likelihood is identical at both values (-110.832 for groups 1–3), so β is not identified without restricting u.

Results to note. $\beta = .9999$ fits better than $\beta = 0$ in the pooled sample (LR 12.782, $p = .0035$) and in groups 1–3 ($p = .029$), not significantly in group 4 ($p = .053$) (Table IX). Conditional independence is rejected for groups 1–3 ($p = .0043$) but not group 4 (Table XI).

Assumptions: stated and implicit.

- *Stated by the author:* additive i.i.d. EV1 shocks; CI; Markov mileage; infinite-horizon stationary problem; β fixed; independence across buses.
- *Implicit, in our reading:* no persistent unobserved differences within a bus group (the CI rejection suggests otherwise); the agent knows the transition law; replacement is always affordable (no budget constraint); one asset per decision; behavior stable over ten years.

A.1.2 Schiraldi (2011): automobile replacement with transaction costs

Why it is included: it is the car-specific version of Rust's keep-or-replace engine, with a resale market, a per-purchase transaction cost τ_{jt}, and a purchase probability that collapses the vehicle menu into one dynamic inclusive value.

Setup. Aggregate registration shares for the Province of Isernia, Italy, 1994–2004, by product (age, engine size, fuel, manufacturer group; 2,178 product-years, p. 268). Annual periods, $\beta = 0.9$ fixed (p. 276). No household choices are observed: "consumer i" is a simulated type, 80 Halton draws of income y_i and age taste υ_i (p. 279). *Given state:* held car $k \in J_{t-1} \cup \{0\}$ and market information Ω_{it} (attributes, prices, availability), seen by consumer and researcher; ξ_{jt}, τ_{jt}, and shocks ϵ_{ijt}, seen by the consumer only. *Actions:* keep k, sell and buy $j \in J_t$ (new or used), or sell and hold no car ($j = 0$, $p_{0t} = 0$). At most one car per consumer (p. 272).

Equations and derivation.

1. **Flow utilities** (1)–(3), pp. 272–273: keep $\tilde{u}_{it}^{k} = x_{kt}\alpha_i^x + \xi_{kt} + \epsilon_{ikt}$; buy $u_{it}^{kj} = x_{jt}\alpha_i^x + \xi_{jt} - \alpha_i^p p_{jt} - \tau_{jt} + \alpha_i^p p_{kt} + \epsilon_{ijt}$; exit $u_{it}^{k0} = \alpha_i^p p_{kt} + \epsilon_{i0t}$. Heterogeneity: $\alpha_i^p = \alpha^p/y_i$, $\alpha_i^{age} = \alpha^{age} + \upsilon_i\sigma_{\alpha^{age}}$ (pp. 276–277). *Assumptions used:* only the buyer pays τ; the seller receives p_{kt} in full; full information, no adverse selection (p. 272).

2. **Bellman equation** (4), p. 273: $\hat{V}_i(k, \epsilon_{it}, \Omega_{it})$ is the maximum of the keep branch, the best buy branch $\max_j u_{it}^{kj} + \beta\mathbb{E}[\hat{V}_i(j,\cdot) \mid \Omega_{it}, \epsilon_{it}]$, and $1[k \neq 0]$ times the exit branch. Its state is too large to solve (p. 273).

3. **Price shift and rental price** (5)–(6), pp. 273–274. Set $V_i(k,\cdot) \equiv \hat{V}_i(k,\cdot) - \alpha_i^p p_{kt}$ and

$$\phi_{ijt} \equiv x_{jt}\alpha_i^x + \xi_{jt} - \alpha_i^p(p_{jt} - \beta\mathbb{E}_t[p_{jt+1}]), \qquad \phi_{i0t} = 0.$$

   The bracket is a one-year rental price. The shift cancels the resale term in (2)–(3); in our rewriting, the branches of (5) become exit $\epsilon_{i0t} + \beta\mathbb{E}[V_i(0)]$, buy $\phi_{ijt} - \tau_{jt} + \epsilon_{ijt} + \beta\mathbb{E}[V_i(j)]$, keep $1[k \neq 0](\phi_{ikt} + \epsilon_{ikt} + \beta\mathbb{E}[V_i(k)])$. *Assumptions used:* no selling cost; prices are in Ω_{it}.

4. **Inclusive value** (7), p. 275, with $EV_i(j, \Omega_{it}) = \int V_i(j, \epsilon_{it}, \Omega_{it})\, dP_\epsilon$ (p. 274):

$$\delta_{it} = \ln\Big(\textstyle\sum_{j\in J_t} \exp\big(\phi_{ijt} - \tau_{jt} + \beta\mathbb{E}[EV_i(j, \Omega_{it+1}) \mid \Omega_{it}]\big)\Big).$$

   Assumptions used: i.i.d. type-I extreme-value shocks and Rust's conditional independence (pp. 273–274). Unlike static versions, δ_{it} contains future decisions (p. 275).

5. **State reduction** (Assumption 1, (8)–(9), p. 275): $G_i(\phi_{ikt+1}, \delta_{it+1} \mid \Omega_{it}) = G_i(\phi_{ikt+1}, \delta_{it+1} \mid \phi_{ikt}, \delta_{it})$, estimated as $\delta_{it+1} = \rho_{1i} + \rho_{2i}\delta_{it} + \eta_{it}$ and $\phi_{ijt+1} = \gamma_{1i} + \gamma_{2i}\phi_{ijt} + \gamma_{3i}\delta_{it} + \mu_{it}$, independent normal errors, type-specific parameters. *Assumptions used:* Markov beliefs, called bounded rationality and possibly inconsistent with any supply model (p. 275); policy is not a state (fn. 14).

6. **Integrated Bellman equation** (10), p. 276: $EV_i(\phi_{ikt}, \delta_{it}) = \ln\big(\exp(\delta_{it}) + 1[k \neq 0]\exp(W_{it}^k) + \exp(W_{it}^0)\big)$, with, in our shorthand, keep value $W_{it}^k = \phi_{ikt} + \beta\mathbb{E}[EV_i(\phi_{ikt+1}, \delta_{it+1}) \mid \phi_{ikt}, \delta_{it}]$ and exit value $W_{it}^0 = \beta\mathbb{E}[EV_i(0, \delta_{it+1}) \mid \delta_{it}]$. *Check:* (10) conditions the exit term on $(\phi_{ikt}, \delta_{it})$, (11)–(12) on δ_{it}; these agree because (8) makes δ_{it+1} depend on δ_{it} only.

7. **Choice probabilities** (11)–(12), p. 276, with $D_{it} = \exp(\delta_{it}) + \exp(W_{it}^0) + 1[k \neq 0]\exp(W_{it}^k)$:

$$d_{it}^{kj} = \frac{\exp\big(\phi_{ijt} - \tau_{jt} + \beta\mathbb{E}[EV_i(\phi_{ijt+1}, \delta_{it+1}) \mid \delta_{it}, \phi_{ijt}]\big)}{D_{it}}, \qquad \tilde{d}_{it}^{k} = \frac{\exp(W_{it}^k)}{D_{it}}.$$

   Check (ours): $\sum_{j\in J_t} d_{it}^{kj} = \exp(\delta_{it})/D_{it}$; the market enters timing only through δ_{it}.

8. **Aggregation** (13)–(15), p. 276: $s_{jt}^D = \int \sum_k d_{it}^{kj} s_{ikt-1}\, dP_\upsilon$, $\tilde{s}_{kt}^H = \int \tilde{d}_{it}^{k} s_{ikt-1}\, dP_\upsilon$, $s_{ijt} = \tilde{s}_{ijt}^H + s_{ijt}^D$. Holdings s_{ikt} evolve endogenously by type. *Assumptions used:* market size is the adult population; initial holdings come from a static model without transaction costs (pp. 276, 279).

9. **GMM** (Assumption 2, (16), p. 277): the innovation $\varsigma_{jt} = \xi_{jt} - \lambda\xi_{jt-1}$ satisfies $\mathbb{E}[Z_{jt}\varsigma_{jt}] = 0$, and the estimator is $\hat{\theta} = \arg\min_\theta G(\theta)' W^{-1} G(\theta)$, with nonlinear parameters $\theta_1 = \{\alpha^p, \lambda, \sigma_{\alpha^{age}}\}$. Scrappage take-up in 1997–1998 enters as heavily weighted micromoments (fn. 17). *Assumptions used:* instruments orthogonal to the innovation, not to ξ; observed next-year prices replace $\mathbb{E}_t[p_{jt+1}]$ (p. 276).

10. **Two share inversions** (17)–(18), p. 279, matching each used product's holding and purchase shares:

$$\hat{\phi}'_{jt} = \hat{\phi}_{jt} + \psi_1(\ln \check{s}^H_{jt} - \ln \tilde{s}^H_{jt}(\hat{\phi}_{jt}, \tau_{jt}, \theta)),$$
$$(\hat{\phi}_{jt} - \tau_{jt})' = (\hat{\phi}_{jt} - \tau_{jt}) + \psi_2(\ln \check{s}^D_{jt} - \ln s^D_{jt}(\hat{\phi}_{jt}, \tau_{jt}, \theta)),$$

$\psi_1 = 1 - \beta$, $\psi_2 = (1-\beta)^2$; τ_{jt} follows by difference. Keepers do not pay τ, so holding shares identify $\hat{\phi}_{jt}$ and purchase shares $\hat{\phi}_{jt} - \tau_{jt}$ (pp. 279–280). *Note:* the author calls the argument heuristic (p. 280), with no uniqueness proof (p. 279); new-car τ is not identified (fn. 32).

Results to note. $\alpha_p = -52.37$ (4.64), $\lambda = 0.35$ (0.03) with micromoments; $\alpha_p = -2.04$ (4.52) without transaction costs (Table 4, p. 281). The static price coefficient -9.32 (6.13) is starred as significant at 5% though its t-ratio is about 1.5. Average own-price elasticities: static -3.05, dynamic -4.07 (p. 283); the text says the myopic model "upwardly biases" them, but the static values are smaller in magnitude, so we read it as a bias toward zero. Mean used-car τ, the cost of a hypothetical purchase, falls from about EUR 3200 (1994) to EUR 2000 (2004), without standard errors (p. 284, Fig. 4).

Assumptions: stated and implicit.

- *Stated by the author:* i.i.d. EV1 shocks and conditional independence; one car per consumer; buyer-paid τ; full information; perfectly persistent preferences (pp. 270, 279); Assumptions 1–2; $\beta = 0.9$.
- *Implicit, in our reading:* nothing but τ makes a consumer keep a depreciating car, so unmodeled persistent tastes, liquidity, or selling costs load onto τ; expectations are inconsistent (perfect foresight in ϕ, AR(1) beliefs in EV); no budget constraint; no sensitivity to β reported.

Question about the paper.

- An unobserved persistent keep premium: which way does it bias τ in (17)–(18), and what separates them?

A.1.3 Gavazza and Lanteri (2021): credit shocks and car replacement

Why it is included: it is the only car model in our set with liquid wealth, persistent income risk, and a borrowing limit inside the keep, trade, or scrap decision, so it supplies the budget and feasibility structure Rust-type models lack.

Setup. A continuum of households, annual, one indivisible car each (Sec. 4). *Given state:* bonds b, income w, car status $n \in \{1, \ldots, N+1\}$ ($N+1$ is no car), and a permanent type θ, all observed by household and researcher; income and depreciation shocks realize before decisions. *Actions:* consumption c, bonds b', car to hold $\tilde{n}$ (keep if $\tilde{n} = n$, else trade or scrap). *Prices:* p_1, p_N fixed; used prices and p_b clear markets. No choice shocks anywhere.

Equations and derivation.

1. **Preferences** (1): $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it}, d_{it})$, $\beta \in (0,1)$, with the CRRA–Cobb–Douglas form (Sec. 5, following Berger and Vavra 2015)

$$u(c,d) = \frac{(c^\alpha d^{1-\alpha})^{1-\gamma}}{1-\gamma}, \qquad \alpha = 0.95,\ \gamma = 2.$$

Assumptions used: expected utility, time separability, Cobb–Douglas aggregation. *Note:* $\gamma = 2$ comes from the Aiyagari range, not car data. The credit-shock mechanism runs through precautionary saving, so results depend on γ.

2. **Car services and depreciation** (2): $d(n, \theta_i) = q_n$ for an owner, θ_i for a non-owner. Qualities $q_1 > \cdots > q_N$, new cars q_1; q_n becomes q_{n+1} with probability π_n; q_N is scrapped. Calibration: $N = 4$, $(\pi_1, \pi_2, \pi_3) = (1/3, 0.1, 0.5)$, $(q_1, q_2, q_3) = (1, 0.3, 0.1)$, $\theta \in \{0, 1\}$ with $\mathbb{P}(\theta = 1) = 0.1$. *Assumptions used:* Markov depreciation on four rungs; no horizontal differentiation.

3. **Transaction cost and income.** A seller pays $\lambda(p_n) = \lambda_0 + \lambda_1 p_n$, $\lambda_0 = 0.03$, $\lambda_1 = 0.15$ (affine; no time or search cost). Income: $\log w_{i,t+1} = \rho \log w_{i,t} + \epsilon_{i,t+1}$, $\epsilon \sim N(-0.5\sigma_w^2, \sigma_\epsilon^2)$, $\rho = 0.9$, $\sigma_\epsilon = 0.2$, on a 3-state Rouwenhorst chain. *Check:* the paper states $\sigma_w = \sigma_\epsilon / \sqrt{1 - \rho^2} = 0.63$, but $0.2/\sqrt{0.19} = 0.459$; 0.63 is $0.2/\sqrt{1 - \rho}$.

4. **Borrowing limit** (3): $b_{i,t+1} \geq \phi$, $\phi \leq 0$, on a one-period non-contingent bond priced p_b, real rate $r = 1/p_b - 1$; calibration $\phi = -1$ (borrowing up to mean annual income). Taxes satisfy $b_G(1 - p_b) = \tau$ (4). *Assumptions used:* one limit common to all households; no default, no amortizing loan contract.

5. **Bellman and budget** (5)–(6):

$$V(b, w, n; \theta) = \max_{c, b', \tilde{n}} u\big(c, d(\tilde{n}, \theta)\big) + \beta \mathbb{E}\big[V(b', w', n'; \theta) \mid \tilde{n}, w\big],$$
$$c + p_{\tilde{n}} + p_b b' + \tau = w + p_n - \lambda(p_n)\, \mathcal{I}(\tilde{n} \neq n) + b.$$

Check: if $\tilde{n} = n$, $p_{\tilde{n}}$ and p_n cancel, so keeping entails no vehicle transaction outlay. Written out (our reconstruction), $\mathbb{E}[V \mid \tilde{n}, w] = \sum_{w'} F_w(w, w')[(1 - \pi_{\tilde{n}}) V(b', w', \tilde{n}; \theta) + \pi_{\tilde{n}} V(b', w', \tilde{n} + 1; \theta)]$ for $\tilde{n} \leq N - 1$. Splitting (5) gives car-specific values $v_{\tilde{n}}$ and $V = \max_{\tilde{n}} v_{\tilde{n}}$. *Assumptions used:* $\tilde{n}$, b' chosen jointly after shocks; n' depends only on $\tilde{n}$, w' only on w.

6. **Feasibility bound** (our reconstruction from (3) and (6)): a trade to $\tilde{n}$ is feasible only if $c > 0$ at $b' = \phi$,

$$b > p_{\tilde{n}} - \big[p_n - \lambda(p_n)\big] + p_b \phi + \tau - w.$$

The bracket is trade-in equity; tightening ϕ raises the bound by p_b per unit, as does a lower used price p_n for trade-in owners. *Note:* necessary only; with $\gamma = 2$, $u \to -\infty$ as $c \to 0$, so optimal thresholds lie strictly above it.

7. **Threshold policy** (Fig. 6): the car policy $g_n(b, w, n; \theta)$ is deterministic and switches at wealth thresholds in (w, b). Read from the plot, the $q_2 \to q_1$ threshold falls from about 12 at $w = 0.5$ to about 5 at $w = 1.5$; the q_3 threshold sits at or just above $\phi = -1$. Rich households replace q_2 with q_1, mid-wealth ones scrap q_3 for q_2, indebted low-income ones keep q_3 until they deleverage. *Key gap:* with no taste shocks, $\mathbb{P}(\text{trade} \mid b, w, n, \theta) \in \{0, 1\}$, identical households act identically, and the calibration does not specify a statistical household-choice likelihood. Adding EV1 shocks to each $v_{\tilde{n}}$ would turn the threshold into a logit, as in Rust's (4.11)–(4.14) (Appendix A.1.1).

8. **Collateral variant** (11), Sec. 6.5: $b_{i,t+1} \geq \phi_t\big(\chi_0 + \chi_1 \mathbb{E}_t[p_{n_{i,t+1}} \mid \tilde{n}_{it}]\big)$, with $\chi_1 = 0.85$, $\chi_0 = 0.8$. Borrowing capacity rises with the chosen car's expected resale value; no lender pricing or loan-to-value contract.

9. **Market clearing** (7)–(9): bonds, $\int g_b \, dm = b_G$; new cars, $\int \mathcal{I}(g_n = 1) \, dm = \int dm(b, w, n = 1; \theta) + x$ (sets output x); used $\bar{n} = 2, \ldots, N - 1$, $\int \mathcal{I}(g_n = \bar{n}) \, dm \leq \int dm(b, w, \bar{n}; \theta)$, strict only with scrapping, then $p_{\bar{n}} = p_N$. Calibrated, $p_3 = p_N$, so only p_2, p_b are endogenous.

10. **Calibration** (Table 1): nothing is estimated statistically. Literature: $\gamma, \rho, \sigma_\epsilon$. One target each: α to a 5% vehicle expenditure share; β to a 2.5% real rate; ϕ to 10% constrained households; b_G to liquid assets/GDP of 1.78 ($b_G = 1.5$ does not visibly map to it); π_n to a 15-year lifetime and 10% scrappage at 15; q_2, q_3 to 3-year depreciation; $p_1 = 0.45$ to price/income; $p_N = 0.036$ to residual value; $\mathbb{P}(\theta = 1)$ to the no-car share; λ_0, λ_1 to NADA retail minus trade-in spreads. Targets interact (e.g. π_n, q_n).

Results to note. Tightening ϕ from -1 to -0.4 over three years lowers the real rate to 1.9% and cuts new-car sales and scrappage by about 7 and used prices by about 10 percentage points (Fig. 8). Constrained owners postpone scrapping, p_2 falls, and the upgrade cost $p_1 - [p_2 - \lambda(p_2)]$ rises for unconstrained ones, who also delay (Sec. 6.2); without used-market clearing, new sales rise (Fig. 10). The collateral variant nearly doubles the drop (Fig. 16).

Assumptions: stated and implicit.

- *Stated by the authors:* steps 1–5 and 10; fixed p_1 (relaxed in Sec. 6.4); competitive used markets; annual period.
- *Implicit, in our reading:* no choice shocks (never discussed); no default or loan schedule; ϕ stands in for credit supply, not a lender's decision; the no-car margin is set by type, not liquidity ($\theta = 0$ gives $u = -\infty$ without a car, $\theta = 1$ gets $d = q_1$ free).

A.2 C2. Identification and estimation

A.2.1 Magnac and Thesmar (2002): identification of dynamic discrete choice

Why it is included: it states exactly which primitives choice and transition data cannot recover, so it tells a modeler what must be fixed before estimated utilities mean anything, and when β can be estimated.

Setup. A theory paper: no data, simulation, or application. Two periods (a short panel), alternatives $i \in I = \{1, \dots, K\}$, with K the reference alternative. *Given state:* x (includes time, fn. 2, p. 802), observed by agent and researcher, with discrete finite supports X, X' (DS, p. 804); shocks $\tilde{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_K)$, observed only by the agent. *Action:* $d = i$. *Data:* $\mathbb{P}(d', x', d \mid x) = \mathbb{P}(d' \mid x')\mathbb{P}(x' \mid x, d)\mathbb{P}(d \mid x)$ (p. 805). The paper calls β the "discount rate"; it is the discount factor.

Equations and derivation.

1. **Bellman equations in deterministic parts** (1)–(2), p. 804. From $v_i(x, \tilde{\varepsilon}) = u_i(x) + \varepsilon_i + \beta\mathbb{E}(\max_j v_j(x', \tilde{\varepsilon}') \mid x, d = i)$, CI removes $\tilde{\varepsilon}$ from the continuation term, so $v_i = v_i^*(x) + \varepsilon_i$ and

$$v_i^*(x) = u_i^*(x) + \beta\mathbb{E}\Big(\max_j (v_j^*(x') + \varepsilon_j') \mid x, d = i\Big).$$

   Assumptions used: AS (pp. 802–803), $u_i = u_i^*(x) + \varepsilon_i$, $\mathbb{E}(\varepsilon_i \mid x) = 0$, G absolutely continuous, utilities otherwise unrestricted; PE (p. 803), beliefs equal true transitions, "quite a strong assumption"; CI (p. 803), $\tilde{\varepsilon}'$ independent of $\tilde{\varepsilon}$ given (x, d), which rules out unobserved heterogeneity (p. 804). Under PE transitions come from data, so the structure is $b = \{u_1^*(X), \dots, u_K^*(X), v_1^*(X'), \dots, v_K^*(X'), G, \beta\}$, with choice probabilities (3), p. 805.

2. **Hotz–Miller inversion** (4), p. 806: given G, $v_i^*(\tilde{x}) = v_K^*(\tilde{x}) + q_i(\tilde{p}(\tilde{x}); G)$, $q_K = 0$; in period 2, (5): $(v_1^*, \dots, v_{K-1}^*)(X') = \varphi(\tilde{p}(X'); v_K^*(X'))$. *Assumption used:* G known (Hotz and Miller, 1993). Choice probabilities give value *differences*, never the level v_K^*.

3. **The function** R, p. 806: $R(z;G) = \mathbb{E}_G \max_i(z_i + \varepsilon_i - \varepsilon_K)$. Adding and subtracting v_K inside the max (Appendix A, pp. 813–814) gives $\mathbb{E}_G(\max_j v_j(x', \tilde{\varepsilon}') \mid x') = v_K^*(x') + R(q(\tilde{p}(x')); G)$. *Assumptions used:* $\mathbb{E}(\varepsilon_K \mid x') = 0$ (AS(i)) and (4) at x'. The continuation value is observed CCPs plus one unknown level.

4. **Lemma 1** (6), p. 806. Substitute step 3 into (2), subtract the Kth equation (Appendix (14)–(15)), and eliminate $v_K^*(x)$. For $i < K$,

$$u_i^*(x) = u_K^*(x) - \beta\big[\mathbb{E}(v_K^*(x') \mid x, d = i) - \mathbb{E}(v_K^*(x') \mid x, d = K)\big] \\ + q_i(\tilde{p}(x); G) - \beta\big[\mathbb{E}(R(q(\tilde{p}(x')); G) \mid x, d = i) - \mathbb{E}(R(q(\tilde{p}(x')); G) \mid x, d = K)\big].$$

   Check: as printed in (6) and (8), the parentheses put the last $\mathbb{E}(\cdot \mid x, d = K)$ term outside β. Appendix (14)–(15) and the estimator $Y_i = q_i(\tilde{p}(x)) - \beta R(q(\tilde{p}(x')))$ (p. 808) imply that β multiplies the whole difference, as written here.

5. **Mapping** (7) **and Proposition 2**, pp. 806–807. (6) is a map

$$(u_1^*, \ldots, u_{K-1}^*)(X) = \phi(\tilde{p}(X), \pi; \beta, G, u_K^*(X), v_K^*(X')),$$

   with $\pi = \mathbb{P}(X' \mid X, I)$; the Kth equation only defines $v_K(X)$ and carries no information. With $C = \{c = (\beta, G, u_K^*(X), v_K^*(X'))\}$: (i) for fixed c, exactly one $(u_1^*, \ldots, u_{K-1}^*)(X)$ is compatible with $\tilde{p}(\tilde{X})$; (ii) for all $c, c' \in C \times C$,

$$\{\phi(.;c), \varphi(.;c), c\} \overset{O}{\Leftrightarrow} \{\phi(.;c'), \varphi(.;c'), c'\}.$$

   Consequence: non-identification is exact; every c, including $\beta = 0$ (Manski's static equivalence, p. 807), fits the data equally well. It is larger if G or β depends on x.

6. **Corollary 3 and the moment condition** (8), p. 807. Rearranging (6), for $i < K$,

$$q_i(\tilde{p}(x); G) - \beta\big[\mathbb{E}(R(q(\tilde{p}(x')); G) \mid x, d = i) - \mathbb{E}(R(q(\tilde{p}(x')); G) \mid x, d = K)\big] = U_i(x),$$

$$U_i(x) \equiv \big[u_i^*(x) + \beta\mathbb{E}(v_K^*(x') \mid x, d = i)\big] - \big[u_K^*(x) + \beta\mathbb{E}(v_K^*(x') \mid x, d = K)\big].$$

   Given β and G the left side is data, so $U_i(X)$ is identified with $v_K^*(X')$ unknown. U_i compares "i now, K tomorrow, then optimal" with "K now and tomorrow, then optimal." *Note:* $U_i = u_i^* - u_K^*$ only if $\beta = 0$; otherwise utility differences need $v_K(X')$. The step 4 parenthesis check applies to (8).

7. **Proposition 4**, p. 809. Restriction: $U_i(x_1) = U_i(x_2)$ for some i and $x_1 \neq x_2$. With G fixed, differencing (8) across x_1, x_2 is linear in β. Our reconstruction (the paper states only linearity), with R short for $R(q(\tilde{p}(x')); G)$:

$$\beta = \frac{q_i(\tilde{p}(x_1); G) - q_i(\tilde{p}(x_2); G)}{\Delta},$$

$$\Delta = \big[\mathbb{E}(R \mid x_1, i) - \mathbb{E}(R \mid x_1, K)\big] - \big[\mathbb{E}(R \mid x_2, i) - \mathbb{E}(R \mid x_2, K)\big].$$

   Assumptions used: G fixed; rank condition $\Delta \neq 0$; more restrictions overidentify. Stationarity of U_i over three periods supplies one (p. 809). *Note:* absorbing states and terminal dates are testable but not identifying; the reference alternative must be restricted at every period (Sec. 4.1).

8. **Unobserved types** (Section 5, pp. 810–812). (9) conditions everything on a constant type θ. Restricted case: $K = 2$, alternative 2 absorbing with $u_2^* = v_2^* = 0$, $\theta \in \{0, 1\}$ with share $r(\theta)$ independent of x, and (10): θ affects neither transitions nor G. Lemma 5 (11) writes observed CCPs as mixtures with Bayes weights $p(x, \theta)r(\theta)/s(x)$ in period 2, which is dynamic selection. Proposition 6: given G, β, two whole functions and $r(1)$ are unidentified. Proposition 7: adding $r(1)$, additivity $u_1^*(x, \theta) = u_1^*(x) + \theta$, a terminal condition, and rank conditions just identifies u_1^*; β and $r(1)$ need two more exclusions.

Results to note. The unidentified set is exactly $(\beta, G, u_K^*(X), v_K^*(X'))$. Given β and G, U_i is "the only structural object that can be identified" without value-function levels (p. 813). G is never identified, even with parametric utilities (fn. 7, p. 810). No finite-sample results (p. 813).

Assumptions: stated and implicit.

- *Stated by the authors:* AS; PE; CI (relaxed in Section 5); DS; two periods, extensions asserted, not shown (p. 804); G independent of x (p. 803); Section 5 uses two types, two alternatives, and heterogeneity only in preferences (fn. 8, p. 811).
- *Implicit, in our reading:* the same choice set in every state and no budget constraint; rank conditions hold in the data at hand; in a stationary infinite-horizon model the Kth equation is a fixed point determining v_K^*, so fixing $u_K^*(X)$ may suffice (our reconstruction, not in the paper).

A.2.2 Hotz and Miller (1993): conditional choice probability estimation

Why it is included: it shows that the continuation values in a dynamic logit can be recovered from observed choice probabilities and transitions, so a purchase-timing model can be estimated without solving the dynamic program at every trial parameter.

Setup. A finite-horizon dynamic discrete-choice problem with J actions per period (Sec. 2, pp. 498–500). *Given state:* the history $H_t = (b_0', b_1, \ldots, b_{t-1})'$ of an initial endowment and past outcomes, observed by agent and researcher; shocks $\varepsilon_t = (\varepsilon_{t1}, \ldots, \varepsilon_{tJ})$, observed only by the agent. *Action:* $d_{tj} \in \{0, 1\}$, $\sum_j d_{tj} = 1$. *Objective:* $\mathbb{E}_0 \sum_{t=0}^{T} \sum_j d_{tj}[u_j^*(H_t) + \varepsilon_{tj}]$, with β^t inside u^*. Application: 2,088 couples, 1975 National Fertility Survey, 10,257 couple-years, choosing among no contraception, temporary contraception, and irreversible sterilization (pp. 513–514).

Equations and derivation.

1. **Payoff and valuation** (2.3), (2.6)–(2.8), pp. 499–500: $u_{tj} = u_j^*(H_t) + \varepsilon_{tj}$; $v_j(H_t)$ is expected optimal future utility after choosing j, excluding the current payoff; $p_k(H_t) = \mathbb{P}\{k = \arg\max_j [u_{tj}^* + \varepsilon_{tj} + v_{tj}] \mid H_t\}$. *Assumptions used:* additive shocks, independent of H_t; shock distribution $G(\varepsilon_t \mid H_t)$ known up to parameters, with a density (2.4); finite horizon T.
2. **CCPs as a function of value differences** (3.1)–(3.4), p. 500: $p(H_t) = Q(\nu(H_t), H_t)$, $\nu = (v_{t1} - v_{tJ}, \ldots, v_{t,J-1} - v_{tJ})'$. Only differences enter. *Assumption used:* action J serves as the reference.
3. **Inversion** (Proposition 1, p. 501; proof pp. 522–523): Q is invertible in ν, so $\nu(H_t) = Q^{-1}(p(H_t), H_t)$. The Jacobian is diagonally dominant (A.5)–(A.9). *Assumptions used:* a density with positive cross-partials (full support). For type-I extreme value shocks, $v_{tj} - v_{tJ} = \ln(p_j/p_J) - (u_{tj}^* - u_{tJ}^*)$, the form of (4.11), p. 504.

4. **Selection correction and expected payoff** (3.7)–(3.8), p. 501: $W_j = \mathbb{E}(\varepsilon_{tj} \mid H_t, d^o_{tj} = 1)$ is a function of p_t; for extreme value, $W_j = \gamma - \ln p_j$ (4.12). Then $U(p_t, H_t) = \sum_k p_{tk}[u^*_{tk} + W_k]$.

5. **Valuation in CCPs** (3.12), p. 502:

$$V_j(H_t, p(H_t)) = \sum_{s=t+1}^{T} \sum_{H_s \in \mathcal{A}_s(H_t)} U_s \, F_{tj} \prod_{r=t+1}^{s-1} \Big(\sum_k p_{rk} F_{rk} \Big) = v_j(H_t).$$

   Assumptions used: finite outcomes and horizon; agents' beliefs equal the transitions F_j in the data.

6. **Terminating action** (3.13)–(3.14), p. 502: if J, once chosen, must be chosen forever,

$$V_j(H_t) = \sum_{H_{t+1}} \Big\{ U_{t+1} + V_J(H_{t+1}) + \sum_{k<J} p_{t+1,k} Q_k^{-1}(p_{t+1}, H_{t+1}) \Big\} F_j(H_{t+1} \mid H_t),$$

   so only one-period-ahead CCPs are needed. Sterilization plays this role in the application. *Note:* as printed, (4.8)–(4.9) reverse the sign of $v_1 - v_2$ relative to (4.11); (4.11) is the consistent version.

7. **Estimator** (5.4)–(5.9), pp. 506–507: first stage, cell frequencies (5.7) or kernels (5.12) for CCPs and transitions; second stage, GMM on $\mathbb{E}(z_{nt} \otimes [d_{nt} - P(H_{nt}, \hat{p}, \theta)]) = 0$. *Assumptions used:* no unobserved states (Assumption 4); stationary environment and independence across agents, so no aggregate shocks (Assumptions 5–6, p. 506); all feasible histories sampled (p. 509).

8. **Asymptotics** (Proposition 2, p. 508): $N^{1/2}$-consistent and asymptotically normal; the covariance (5.11) corrects for the first stage. Kernel first stages are asymptotically biased; a combination of estimators with different bandwidths removes the bias ((B.27), p. 527).

Results to note. The reduced form is precise ($\chi^2 = 220$, 33 d.f., Table IV, p. 517), and the first-stage correction to standard errors was negligible (p. 517). All three structural specifications are rejected ($p = 0.0034$ for the least restrictive, p. 522), and β is imprecise: 0.646 (s.e. 0.678) in Model C (Table VI, p. 521). No computing times are reported.

Assumptions: stated and implicit.

- *Stated by the authors:* additive shocks with known distribution; finite horizon and finite outcomes; no unobserved state variables; no aggregate shocks; independence across agents; serially uncorrelated income shocks (fn. 10).
- *Implicit, in our reading:* no persistent unobserved heterogeneity (a type would make the CCPs mix types); cells are populated well enough to estimate $\ln p$ without large error; counterfactuals stay within the observed regime, since regime changes may require solving the model (p. 503).

A.2.3 Aguirregabiria and Suzuki (2014): what normalizations do to counterfactuals

Why it is included: it applies Magnac and Thesmar's non-identification to a binary dynamic choice whose payoff has several cost components, and then states which counterfactuals survive the normalization a modeler must impose and what normalized estimates actually measure.

Setup. A theory paper with one numerical example; no data. *Decision maker:* a single firm (later an oligopoly game). *Action:* $a_t \in \{0, 1\}$, inactive or active. *Given state:* incumbency $k_t \equiv a_{t-1}$

(endogenous) and exogenous Markov states $\mathbf{z}_t = (\mathbf{z}_t^v, \mathbf{z}_t^c)$ with transition f_z, both observed by firm and researcher; shocks ε_t observed only by the firm. *Data:* many markets, short panels; β, f_z and variable profit vp are treated as known (Sec. 3.1, p. 276). Fixed profit has three unknown functions of $\mathbf{z}^c$: fixed cost fc, entry cost ec, scrap value sv.

Equations and derivation.

1. **Payoffs and choice rule** (3), (5)–(10), pp. 272–274:

$$\Pi_t = \begin{cases} k_t\, sv(\mathbf{z}_t^c) & a_t = 0, \\ vp(\mathbf{z}_t^v) - fc(\mathbf{z}_t^c) - (1-k_t)\, ec(\mathbf{z}_t^c) & a_t = 1, \end{cases}$$

   plus $\varepsilon_t(0) = k_t \varepsilon_t^{sv}$, $\varepsilon_t(1) = -\varepsilon_t^{fc} - (1-k_t)\varepsilon_t^{ec}$. With conditional value $v(a,k,\mathbf{z})$ and $\tilde{v}(k,\mathbf{z}) = v(1,k,\mathbf{z}) - v(0,k,\mathbf{z})$, the firm is active iff $\tilde{\varepsilon}_t \le \tilde{v}(k_t,\mathbf{z}_t)$, so $P(k,\mathbf{z}) = F_{\tilde{\varepsilon}|k}(\tilde{v}(k,\mathbf{z}))$. *Assumptions used:* additive shocks, i.i.d. over time, independent of $(\mathbf{z}_t, k_t)$, zero mean (p. 272); no shock in variable profit (p. 273); f_z independent of a_t; the inactive non-incumbent earns zero, $\pi(0,0,\mathbf{z}) = 0$ ((3), p. 272).

2. **CCP inversion** (16), p. 278: $\tilde{v}(k,\mathbf{z}) = F^{-1}_{\tilde{\varepsilon}|k}(P(k,\mathbf{z}))$. *Assumptions used:* $F_{\tilde{\varepsilon}|k}$ strictly monotone and identified by Proposition 1 (p. 277), which needs a *finitely lived* firm and vp_t as a special regressor with continuous support; no infinite-horizon proof is known to the authors (fn. 13). Without price data everything is identified only up to an unknown scale (pp. 277–278).

3. **Continuation difference in observables** (17)–(19), pp. 278–279. With $\bar{V}(k,\mathbf{z}) = \int V\, dG$, (18) gives $\bar{V}(k,\mathbf{z}) = v(0,k,\mathbf{z}) + S(\tilde{v}(k,\mathbf{z}), F_{\tilde{\varepsilon}|k})$, $S(\tilde{v},F) \equiv \int_{-\infty}^{\tilde{v}} [\tilde{v} - \tilde{\varepsilon}]\, dF(\tilde{\varepsilon})$, a known function. Since both $(a,k) = (0,1)$ and $(0,0)$ lead to $k = 0$ next period, $v(0,1,\mathbf{z}') - v(0,0,\mathbf{z}') = sv(\mathbf{z}^{c\prime})$ (p. 279). *Assumptions used:* $\mathbb{E}\varepsilon(0) = 0$; $\mathbf{z}$ exogenous. Our reconstruction of the flow difference: $\pi(1,k,\mathbf{z}) - \pi(0,k,\mathbf{z}) = vp - fc - ec + k(ec - sv)$.

4. **The identified function** Q (20), p. 279. Subtracting the known S terms from $\tilde{v}$ defines $Q(k,\mathbf{z})$, and

$$Q(k,\mathbf{z}) = vp(\mathbf{z}^v) - [fc(\mathbf{z}^c) + ec(\mathbf{z}^c)] + k\,[ec(\mathbf{z}^c) - sv(\mathbf{z}^c)] + \beta \textstyle\sum_{\mathbf{z}^{c\prime}} f_z(\mathbf{z}^{c\prime} \mid \mathbf{z})\, sv(\mathbf{z}^{c\prime}).$$

   Q is "active today, exit next period" minus "exit today" (p. 279). In our reading it is Magnac and Thesmar's $U_i(x)$ (their (8); step 6 of Section A.2.1) with "inactive" as reference.

5. **Proposition 2: non-identification** (21), pp. 279–280; proof p. 301. Two equations ($k = 0, 1$) per $\mathbf{z}$ in three functions identify only

$$ec - sv = Q(1,\mathbf{z}) - Q(0,\mathbf{z}), \qquad fc + ec - \beta \textstyle\sum f_z\, sv(\mathbf{z}^{c\prime}) = vp(\mathbf{z}^v) - Q(0,\mathbf{z}).$$

   For any $\lambda(\mathbf{z}^c)$, $(ec{+}\lambda,\ sv{+}\lambda,\ fc{-}\lambda{+}\beta\mathbb{E}[\lambda(\mathbf{z}^{c\prime}) \mid \mathbf{z}])$ fits the same data. *Check* (our reconstruction): this is the payoff shift $k\lambda(\mathbf{z}) - \beta a \mathbb{E}[\lambda(\mathbf{z}') \mid \mathbf{z}]$ with $V^* = V + k\lambda$, the case $\Phi = k\lambda$ of adding $\Phi(\text{state}) - \beta\mathbb{E}[\Phi(\text{next state}) \mid \text{state, action}]$ to every payoff, which leaves all CCPs unchanged for fixed β and transitions.

6. **What normalized estimates measure** (22) and Table 1, pp. 280–281. Under $\widehat{sv} = 0$: $\widehat{ec} = ec - sv$ (ex-ante sunk entry cost) and $\widehat{fc} = fc + sv - \beta\mathbb{E}[sv(\mathbf{z}_{t+1}^c) \mid \mathbf{z}_t]$. Under $\widehat{fc} = 0$: $\widehat{ec}$ and $\widehat{sv}$ each add $\sum_{r\ge 0} \beta^r \mathbb{E}[fc(\mathbf{z}_{t+r}^c) \mid \mathbf{z}_t]$, the present value of operating forever (p. 281). Under $\widehat{ec} = 0$: $\widehat{sv} = sv - ec$, $\widehat{fc} = fc + ec - \beta\mathbb{E}[ec(\mathbf{z}_{t+1}^c) \mid \mathbf{z}_t]$. *Check* (our reconstruction of row 2): (22) gives the recursion $\widehat{sv} - sv = fc + \beta\mathbb{E}[(\widehat{sv} - sv)']$, a contraction for $\beta < 1$, bounded fc.

7. **Counterfactuals: Lemma 1 and Propositions 3–5**, pp. 284–286; proofs pp. 300–302. Lemma 1: for given (β, f_z) the map from the whole CCP vector to the whole Q vector, (23), is one-to-one, so P^* is identified iff Q^* is; in our reading, the Jacobian's nonsingularity is asserted without the full argument. Split $\mathbf{z} = (\mathbf{z}^{nosv}, \mathbf{z}^{sv})$, $\mathbf{z}^{sv}$ the states entering sv. Differencing (20) gives (24), printed as $Q^* = Q^0 + \text{RHS}$; with $\Delta_Q \equiv Q^* - Q^0$ (our notation),

$$\Delta_Q(k, \mathbf{z}) = \Delta_{vp} - [\Delta_{fc} + \Delta_{ec}] + k[\Delta_{ec} - \Delta_{sv}] + \beta^0 \sum f^0_{z,sv}\, \Delta_{sv}(\mathbf{z}^{sv\prime}) + \beta^0 \sum \Delta_{f_{z,sv}}\, [sv^0 + \Delta_{sv}](\mathbf{z}^{sv\prime}) + \Delta_\beta \sum [f^0_{z,sv} + \Delta_{f_{z,sv}}]\, [sv^0 + \Delta_{sv}](\mathbf{z}^{sv\prime}).$$

Prop. 3 (p. 285): with $\Delta_\beta = 0$, $\Delta_{f_{z,sv}} = 0$ and known $\{\Delta_{vp}, \Delta_{fc}, \Delta_{ec}, \Delta_{sv}, \Delta_{f_{z,nosv}}\}$, the second line vanishes and P^* is identified. Prop. 4 (p. 285): if $\Delta_\beta \neq 0$ or $\Delta_{f_{z,sv}} \neq 0$, Δ_Q depends on the unidentified level sv^0, (30), even with factual and counterfactual (β, f_z) known. Prop. 5 (p. 286): using $\widehat{sv}^0$ for sv^0, the bias (25) is the second line with $[\widehat{sv}^0 - sv^0]$ in place of $[sv^0 + \Delta_{sv}]$, zero iff $\Delta_\beta = \Delta_{f_{z,sv}} = 0$. *Check:* (25) sums over $\mathbf{z}^{sv\prime}$, valid when $\widehat{sv}^0 - sv^0$ depends only on $\mathbf{z}^{sv}$, as under $\widehat{sv} = 0$; under $\widehat{fc} = 0$ the gap depends on every state predicting future fc, so the "innocuous" class is narrower than stated, although the text allows normalizations "not necessarily" $\widehat{sv}^0 = 0$.

8. **Identifying conditions** (Sec. 5, pp. 293–296). Bounds from $ec, fc, sv \geq 0$ and $sv \geq \beta \mathbb{E} sv'$ (Sec. 5.1, p. 293); *Check:* printed $sv_{LOW} = -ec_{LOW} \leq 0$ is weaker than the imposed $sv \geq 0$. Prop. 6 (p. 294): if $\mathbf{z}_2^c$ is excluded from fc and from one of ec, sv, and shifts the transition of $\mathbf{z}_1^c$ with rank $|\mathcal{Z}_1^c| - 1$, the three functions are identified up to a constant; the exclusion is untestable (p. 295). Prop. 7 (p. 295): transition counterfactuals then identified; β-counterfactuals still not. Prop. 8 (p. 296): with acquisition-only entry/exit at a Nash-bargained price (26), classical price error and known bargaining power α, (27) identifies the components; fn. 19 expects selection to make the data reject this.

Results to note. Calibrated land-price example (Sec. 4.3, pp. 287–293): giving high-price markets the low-price process truly shifts entry and stay schedules *down*; under $\widehat{sv} = 0$ they shift *up* (Fig. 4, p. 292), and a potential entrant's unconditional probability of being active is overstated by 4.7 percentage points (15.1 vs. 10.4), p. 292. One illustration chosen to show bias, not evidence of typical magnitude. Table 2 (p. 286): the only transition-change applications are Collard-Wexler (2013) and Das et al. (2007), both under $sv = 0$. *Check*, arithmetic slips in print: Example 2's $\widehat{ec} = 0$ case omits the intercept $(1 - \beta) ec_0$ (p. 282); "$5.6 + 0.4z^c$" should be $5.6 + 0.04z^c$ (p. 289); the stationary s.d. of the stated AR(1) is ≈ 1.15, not 1.3 (p. 288).

Assumptions: stated and implicit.

- *Stated by the authors:* additive i.i.d. shocks independent of $(\mathbf{z}, k)$, no variable-profit shock (pp. 272–273); β, f_z known and vp identified from price/quantity data (p. 276); shock distribution identified only for a finitely lived firm (Prop. 1, fn. 13); $\pi(0, 0, \mathbf{z}) = 0$; large number of markets, short T (p. 276); serial correlation has no "substantive influence" (pp. 272–273), asserted, not shown; the game adds only over-identifying restrictions (pp. 282–283).

- *Implicit, in our reading:* beliefs equal the true f_z; a counterfactual change must be known in utility units and stated in levels, not as a percentage of a normalized component; persistent heterogeneity would need the argument type by type with the type mixture identified elsewhere (Kasahara and Shimotsu, 2009); Prop. 1's finite-horizon restriction leaves the infinite-horizon results without a proved money anchor.

Question for discussion. If replacing a vehicle is read as a simultaneous exit (resale, the analog of sv) and entry (outlay, the analog of ec), does a purchase-outlay counterfactual that also changes the post-purchase liquidity transition fall in Proposition 4's non-identified class?

A.2.4 Arcidiacono and Miller (2020): identification and counterfactuals off short panels

Why it is included: it characterizes every payoff vector that fits the same choice data, shows what a *known* (not normalized) payoff buys, and states which counterfactuals are identified without one, which is the question any counterfactual from a purchase-timing model must answer.

Setup. A theory paper with one two-period example; no data. *Decision maker:* a single agent with horizon $T \leq \infty$; the data end at S, and a "short panel" means $S < T$ in a nonstationary model (p. 473). *Action:* $j \in \{1, \ldots, J\}$, indicators d_{jt}, mutually exclusive (Assumption 1). *Given state:* discrete $x_t \in \{1, \ldots, X\}$, observed by agent and researcher, with transitions $f_{jt}(x_{t+1} \mid x_t)$ (F_j when stationary); shocks ϵ_t with density g, seen only by the agent. *Data:* CCPs $p_{jt}(x)$ for $t \leq S$. T, β, f, g are known (p. 476).

Equations and derivation.

1. **Problem and CCP correction** (1)–(4), pp. 475–476. Conditional value (3): $v_{jt}(x_t) = u_{jt}(x_t) + \beta \sum_{x_{t+1}} V_{t+1}(x_{t+1}) f_{jt}(x_{t+1} \mid x_t)$; $\psi_{jt}(x) \equiv V_t(x) - v_{jt}(x)$, identified from CCPs given g; choose j iff $\epsilon_{kt} - \epsilon_{jt} \leq \psi_{kt} - \psi_{jt}$ for all k (4). The paper states $\psi_{jt} = -\ln p_{jt}$ for standard EV1 (pp. 476, 480). *Assumptions used:* additive, conditionally independent shocks with continuous support; bounded payoffs (Assumptions 2–3, p. 475). *Check:* $-\ln p$ holds for mean-zero EV1; with mean γ, $\psi = \gamma - \ln p$, and the constant cancels wherever ψ enters through differences.

2. **CCP representation** (5)–(6), p. 476, and (10), p. 477. Replacing V_{t+1} by $v_{l,t+1} + \psi_{l,t+1}$ for a normalizing action $l(x, \tau)$ and repeating,

$$v_{jt}(x_t) = u_{jt}(x_t) + \sum_{\tau=t+1}^{T} \sum_{x} \beta^{\tau-t} \left[u_{l(x,\tau),\tau}(x) + \psi_{l(x,\tau),\tau}(x) \right] \kappa^*_{\tau-1}(x \mid x_t, j),$$

   with κ^* the state distribution after j at t and l thereafter; (10) is the case $l = 1$. *Assumptions used:* the definition of ψ and known f. In our reconstruction, nothing uses optimality of l, so any action may serve.

3. **Theorem 1 and Corollary 2: the equivalence class** (7)–(9), pp. 476–477; proofs pp. 481–483. For given (T, β, f, g), choosing in each (x, t) a normalizing action and an arbitrary bounded payoff $c_t(x)$ for it generates exactly the observationally equivalent payoffs. Stationary, (9):

$$u^*_j = u_j + [\mathcal{I} - \beta F_j][\mathcal{I} - \beta F_l]^{-1}(c - u_l).$$

   With $h \equiv [\mathcal{I} - \beta F_l]^{-1}(c - u_l)$ this is $u^*_j = u_j + h - \beta F_j h$ (our rewriting), so $V^* = V + h$ and CCPs are unchanged. One free payoff per state (and period). The usual reading of normalized payoffs as "net of" action 1's is "grossly misleading, if not false" outside a static model (p. 477): they also absorb differences in action 1's future payoffs along the two paths.

4. **Theorem 3: inversion off long panels** (11)–(12), pp. 477–478; proof p. 483. If $u_{1t}(x)$ is *known* for all (x, t), differencing (10) for j and 1 gives

$$\begin{aligned} u_{jt}(x) = u_{1t}(x) + \psi_{1t}(x) - \psi_{jt}(x) \\ + \sum_{\tau=t+1}^{T} \sum_{x_\tau} \beta^{\tau-t} \left[u_{1\tau}(x_\tau) + \psi_{1\tau}(x_\tau) \right] \left[\kappa_{\tau-1}(x_\tau \mid x, 1) - \kappa_{\tau-1}(x_\tau \mid x, j) \right]. \end{aligned}$$

   Check: the page prints $\psi_{1t}(x_\tau)$ inside the sum; the proof and (16) use $\psi_{1\tau}(x_\tau)$. Stationary (12) is printed as $u_j = \Psi_j - \Psi_1 - u_1 + \beta(F_1 - F_j)[\mathcal{I} - \beta F_1]^{-1}(\Psi_1 + u_1)$; the first three terms have the

wrong sign. The proof (p. 483) derives $u_j = u_1 + \Psi_1 - \Psi_j + \beta(F_1 - F_j)[\mathcal{I} - \beta F_1]^{-1}(u_1 + \Psi_1)$, which our numerical check reproduces and printed (12) does not. The systems are exactly identified; plug-in estimators are claimed asymptotically efficient (argued, not proved), and "false restrictions, such as adopting convenient functional forms for the payoffs, typically create misspecifications" (p. 478).

5. **Single-action finite dependence** (13), p. 478. If repeating action 1 makes the state distribution independent of the initial action after ρ periods (terminal actions, renewal, multiperiod renewal), the sum in step 4 stops at $t+\rho$, and "if (13) holds and $u_{1t}(x_t)$ is known for all $t \leq S$, then $u_{jt}(x_t)$ is identified for all $t \leq S - \rho$." The payoff must be "actually known, not just normalized." *Check:* read literally, (13) compares states after $\rho - 1$ repetitions and would exclude the bus-engine renewal the text includes; we read ρ as the number of repetitions.

6. **Corollary 4: short panels** (14), p. 479. Using the known u_1 only up to S,

$$u_{jt}(x) = \ldots + \beta^{S-t} \sum_{x_{S+1}} V_{S+1}(x_{S+1}) \big[\kappa(x_{S+1} \mid x, 1) - \kappa(x_{S+1} \mid x, j)\big],$$

   where "..." is step 4's expression with the sum truncated at S. Flows are linear in the unknown V_{S+1}, which "depends on CCPs that occur after the sample ends"; in general X further restrictions are needed. *Check:* the κ difference sums to zero, so a constant in V_{S+1} drops and $X - 1$ suffice; the exponent β^{S-t} is as printed.

7. **Theorem 5: temporary payoff innovations**, pp. 479–480; proof pp. 483–484. If $\Delta_{jt}(x) = \tilde{u}_{jt}(x) - u_{jt}(x) = 0$ for $t > S$, then $\tilde{p}_{S+1} = p_{S+1}$, $\tilde{p}_{jS}$ follows from $\psi_{kS} - \psi_{jS}$ shifted by $\Delta_{jS} - \Delta_{kS}$, and backward for $t < S$,

$$\begin{aligned}\tilde{\psi}_{kt} - \tilde{\psi}_{jt} = \psi_{kt} - \psi_{jt} + \Delta_{jt} - \Delta_{kt} \\ + \sum_{\tau=t+1}^{S} \sum_{x_\tau} \beta^{\tau-t} \big[\Delta_{1\tau} + \tilde{\psi}_{1\tau} - \psi_{1\tau}\big](x_\tau) \big[\kappa_{\tau-1}(x_\tau \mid x, j) - \kappa_{\tau-1}(x_\tau \mid x, k)\big].\end{aligned}$$

   Only Δ_1 enters, never u_1, so the counterfactual needs no known payoff. The same holds for temporary changes in g (fn. 6). Permanent policies off short panels are not identified (p. 479).

8. **Transition innovations** (15), Sec. 4.2, p. 480. With counterfactual $\tilde{\kappa}$, the $u_{1\tau}$ terms no longer cancel because the paths' state weights change: the counterfactual "cannot be derived without knowing the true systematic payoff for one of the choices, regardless of the sample length." Off short panels it also needs single-action finite dependence; otherwise $\tilde{p}_{jt}$ is "not identified for any t." Argued in text, without a numbered theorem. Our stationary reconstruction: a data-consistent $u_j^* = u_j + h - \beta F_j h$ evaluated under $\tilde{F}_j$ carries the extra payoff $\beta(\tilde{F}_j - F_j)h$, which vanishes when $\tilde{F} = F$. For a given h, this condition is sufficient, not necessary: the exact condition is $(\tilde{F}_j - F_j)h = 0$ for each action. For example, a constant h is preserved by any stochastic transition matrix. Equality of the matrices is necessary if vanishing is required for every unrestricted h (with $\beta > 0$). The paper attributes the stationary payoff-versus-transition result to Aguirregabiria (2005, 2010) and Norets and Tang (2014) (pp. 474, 479).

Results to note. Smoking example (Sec. 5, pp. 480–481), $u_{1t}(1) = 0$, $u_{1t}(2) = c$: normalizing $u_{1t}^* \equiv 0$ biases the smoking payoff by $\beta\pi c$ (17); a subsidy Δ to the sick gives $\tilde{p}_{11}(1) = p_{11}(1)/\{p_{11}(1) + [1 - p_{11}(1)]\exp(\beta\Delta\pi)\}$ from period-1 CCPs alone; a change $\pi \to \tilde{\pi}$ under the normalization is off by the factor $\exp[\beta(\pi - \tilde{\pi})c]$. *Check:* the subsidy is paid at $t = 2 > S = 1$, so it lies outside Theorem 5's literal hypothesis (the result holds because the payment is common to both actions); the printed

base $[p_{12}(1)/p_{12}(2)]$ should be inverted, which our numerical check confirms. Conclusion (p. 481): estimating utilities "purely as a vehicle for making counterfactual predictions is not compelling" without outside knowledge of the utility functions. No finite-sample results.

Assumptions: stated and implicit.

- *Stated by the authors:* Assumptions 1–3 (p. 475): mutually exclusive actions, finite discrete states, additive conditionally independent shocks with continuous support, bounded payoffs; T, β, f, g known, g "extreme value for tractability" (p. 476, fns. 3–4); short-panel counterfactuals restricted to temporary innovations (p. 479); transition counterfactuals need a payoff known from outside the data (Sec. 4.2); persistent heterogeneity left to other work (p. 474), e.g. Kasahara and Shimotsu (2009).
- *Implicit, in our reading:* CCPs are known in every sampled (x, t) cell, so precision with sparse cells is not addressed; beliefs equal realized transitions; how beliefs about aggregate states enter is not specified (p. 479 only excludes them); in a stationary model short histories matter through state coverage and initial conditions (Section A.5.1), not through Corollary 4.

Question for discussion. In a stationary keep/replace model where replacement lowers rather than resets a liquidity state, replace is not a renewal action; is a purchase-outlay counterfactual identified without a known payoff only when it shifts the replace payoff by a known amount in utils and leaves the liquidity transition unchanged?

A.2.5 Kasahara and Shimotsu (2009): identifying finite mixtures of dynamic discrete choices

We work from the authors' working paper (Queen's Economics Department Working Paper No. 1092, which also covers estimation), not the published article, so equation, proposition, remark and page numbers are working-paper locations, and numbering and conditions may differ from the 2009 article. *Why it is included:* it gives sufficient conditions under which permanent unobserved types, their initial distributions, and their choice probabilities are identified nonparametrically from a short panel, the result usually invoked to justify a finite-type household model.

Setup. Theoretical econometrics, with a series-logit estimator (Sec. 4) and a Monte Carlo on a Rust bus model (Sec. 5). M types, known (p. 25), with probabilities π^m; the type is unobserved by the researcher. *Observed:* state $x_t \in X$ and action $a_t \in A = \{0, 1\}$, $t = 1, \ldots, T$, for a random sample of units, $N \to \infty$, T fixed (p. 6). No utilities, β or shocks are used; a forward-looking structural model is a special case (Example 2, (6)–(7), p. 7). Identification is up to relabelling of types (p. 8).

Equations and derivation.

1. **Mixture** (3), p. 6:

$$P(\{a_t, x_t\}_{t=1}^T) = \sum_{m=1}^{M} \pi^m p^{*m}(x_1, a_1) \prod_{t=2}^{T} f^m(x_t \mid x_{t-1}, a_{t-1})\, P^m(a_t \mid x_t, a_{t-1}).$$

   Assumptions used: finite M; the type-specific initial distribution $p^{*m}(x_1, a_1)$ is unrestricted and need not be stationary (Remark 1, p. 6), which is the setting of Heckman (1981) (Example 1, p. 7).

2. **Baseline model and division by transitions** (8)–(9), p. 8. Assumption 1: (a) $P^m(a_t \mid x_t, a_{t-1}) = P^m(a_t \mid x_t)$; (b) $f^m(x' \mid x, a) > 0$ for all (x', x, a); (c) $f^m = f$; CCPs are time-invariant. Then f is identified from observed (x', x, a) and

$$\tilde{P}(\{a_t, x_t\}_{t=1}^T) \equiv \frac{P(\{a_t, x_t\}_{t=1}^T)}{\prod_{t=2}^T f(x_t \mid x_{t-1}, a_{t-1})} = \sum_{m=1}^M \pi^m p^{*m}(x_1, a_1) \prod_{t=2}^T P^m(a_t \mid x_t),$$

a mixture of products: given type, periods are independent. *Assumptions used:* (c) lets f leave the sum; (b) makes the division defined for every sequence.

3. **Submodels** (10)–(11), p. 9: dropping periods gives the same form over any subset $\mathcal{I}$ of periods, with p^{*m} if $1 \in \mathcal{I}$ and without it otherwise. *Our reconstruction:* a period $t \geq 2$ is removed by summing $\tilde{P}$ over a_t at a fixed, arbitrary x_t, which requires (b).

4. **Moments and matrices** (12)–(16), pp. 9–11. Fix $a_t = 1$: $\lambda_\xi^m = P^m(a = 1 \mid x = \xi)$, $\lambda_k^{*m} = p^{*m}((a_1, x_1) = (1, k))$; moments such as $F_{x_2,x_3} = \sum_m \pi^m \lambda_{x_2}^m \lambda_{x_3}^m$ and $F^*_{x_1,x_2,x_3} = \sum_m \pi^m \lambda_{x_1}^{*m} \lambda_{x_2}^m \lambda_{x_3}^m$ fill two $M \times M$ matrices P, P_k with $x_1 = k$, $x_2, x_3 \in \{\xi_1, \ldots, \xi_{M-1}\}$, and a first row and column from the submodels that drop a period. Proof (p. 31): $P = L'VL$ and $P_k = L'VD_kL$, with L the $M \times M$ matrix with rows $(1, \lambda_{\xi_1}^m, \ldots, \lambda_{\xi_{M-1}}^m)$, $V = \operatorname{diag}(\pi^m)$, $D_k = \operatorname{diag}(\lambda_k^{*m})$.

5. **Proposition 1** (pp. 9–10; proof pp. 31–32). If $T \geq 3$, L is nonsingular for some $\xi_1, \ldots, \xi_{M-1} \in X$, and some $k \in X$ has λ_k^{*m} distinct across types, then $\{\pi^m, \lambda_\xi^{*m}, \lambda_\xi^m\}$ is uniquely determined. Construction: $P^{-1}P_k = L^{-1}D_kL$, so the eigenvalues are λ_k^{*m} and the eigenvectors are the columns of L^{-1} up to scale; the column of ones in L fixes the scale; the first column of $(L')^{-1}P = VL$ is $(\pi^1, \ldots, \pi^M)'$; one good set of $M-1$ points gives λ_ζ^m at every other ζ via (29), p. 32 (Remark 2.3). *Assumptions used:* L nonsingular means "different types must respond to [the state's] changes differently" (Remark 2.1, p. 10); for $M = 2$, $\det L = \lambda_{\xi_1}^2 - \lambda_{\xi_1}^1$. Distinct eigenvalues make the eigenvectors unique up to scale.

6. **Corollary 1** (p. 11), the data version: P full rank and $P^{-1}P_k$ with distinct eigenvalues, checkable from sample moments.

7. **Panel length and number of types.** $T \geq 3$ is needed without further restrictions: at $T = 2$ parameters outnumber restrictions (p. 25). $T \geq 4$ if initial distributions do not differ by type (Remark 2.2, p. 10). $M \leq |X| + 1$ under Proposition 1 (Remark 2.4). Proposition 2 (p. 12): odd $T \geq 3$, $u = (T-1)/2$, $\sum_{l=0}^u \binom{|X|+l-1}{l} \geq M$ via Hadamard products.

8. **Extensions** (Sec. 3). Time-varying CCPs: $T \geq 3$ with L_t nonsingular every period (Proposition 3, p. 14). Type-specific transitions: with $Q^m(s_t \mid s_{t-1}) = f^m(x_t \mid x_{t-1}, a_{t-1})P^m(a_t \mid x_t)$, fixing $s_t = \bar{s}$ in odd periods restores a product, requiring $T \geq 6$ and $G_{\bar{s}}$ nonsingular, $\gamma_{\bar{s}}^m(s) = Q^m(\bar{s} \mid s)Q^m(s \mid \bar{s})$ (Proposition 5, p. 16). Lagged choice in the CCP: $T \geq 6$ (Sec. 3.3, p. 19). An extra state with type-common full-support transitions: $T \geq 4$ (Proposition 7, p. 18). Limited transitions: only prefixes are available; Proposition 8 ($T = 4$, pp. 22–23) uses all-$a_t = 0$ paths through sets with $B^h \subseteq \Gamma(0, \{h\})$, $C^h \subseteq \Gamma(0, B^h) \cap \Gamma(0, \{h\})$, $\{k\} \subseteq \Gamma(0, C^h) \cap \Gamma(0, B^h) \cap \Gamma(0, \{h\})$, where $\Gamma(a, W) = \bigcap_{x \in W} \{x' : f(x' \mid x, a) > 0\}$ (Assumption 4(b), p. 22). Local identification by full column rank of the Jacobian (Proposition 10, p. 25). *Check:* Proposition 3 defines ξ_j^t for $t \leq T-1$ but requires L_t for $t \leq T$ (p. 14); Proposition 9's "$T \geq 4$ is odd" means $T \geq 5$ (p. 23).

9. **Estimator** (Sec. 4, pp. 25–28): series logit for P^m; Lemma 1 gives CCP rate $N^{-1/5}$ when $K = N^{1/5}$, $s/r = 4$. *Assumptions used:* continuous compact x; CCPs bounded away from 0 and 1; a *stationary* initial distribution (Assumption 7, pp. 26–27).

Results to note. The theorems are population results with no conditioning analysis. Table 2 (p. 43; two types, $\pi = (0.5, 0.5)$, 100 replications): RMSE of $\hat{\pi}$ at $(N, T) = (500, 3)$ is 0.112 (series) and 0.130 (MLE) against 0.025 with types observed; at $(500, 10)$, 0.041 and 0.043 against 0.024. In our reading, the Monte Carlo design ($T = 3$, limited and, in Table 5, type-specific transitions) lies outside the sufficient conditions of Propositions 5 and 8, so its success rests on the parametric structure.

Assumptions: stated and implicit.

- *Stated by the authors:* finite, known M; Assumption 1(a)–(c); time-invariant CCPs (relaxed in Proposition 3); binary A (multinomial "straightforward," not shown, p. 9); random sampling of complete sequences (p. 6); Assumptions 7–8 for the estimator.
- *Implicit, in our reading:* a balanced panel without type-dependent attrition; a nonsingular L with a determinant near zero satisfies every proposition, so weak identification is invisible; full support excludes deterministic states such as age or tenure; the estimator's stationary initial distribution is a stronger restriction than the unrestricted initial distribution allowed in Remark 1; identifying P^m leaves the Magnac and Thesmar (2002) normalizations to be imposed type by type (fn. 5, p. 5).

Question for discussion. If vehicle age moves deterministically (one month older on keep, zero on replace), Assumption 1(b) fails, the next state reveals the action so the middle-period submodels of step 3 do not exist, and in our reading for the all-keep paths that Proposition 8 uses its set C^h is empty below the age cap: does any result here apply, or would type identification rest on parametric structure?

A.2.6 Higgins and Jochmans (2023): mixtures of Markov chains from four observations

Why it is included: it identifies permanent types, their initial distributions, and type-specific transitions from four consecutive observations without the full-support transitions that Kasahara and Shimotsu (2009) require (Appendix A.2.5), so deterministic state variables are not excluded in form.

Setup. Theoretical econometrics, with an EM estimator and a small Monte Carlo (Appendix B). A latent type $Z \in \{1, \dots, q\}$, q known, with $\mu_z > 0$ (Sec. 2, p. 2). *Observed:* $X_t \in \{1, \dots, r\}$, the choice and state enumerated into one variable (pp. 2, 8). Given $Z = z$, X_t is a first-order, time-homogeneous Markov chain with initial distribution $s_z(x) = \mathbb{P}(X_1 = x \mid Z = z)$, not assumed stationary, and kernel $k_z(x, x') = \mathbb{P}(X_t = x' \mid X_{t-1} = x, Z = z)$. The kernel absorbs choice probabilities, state transitions and lagged-choice dependence, all type-specific and unrestricted. No behavioral model. Identification is up to one common relabelling of types (p. 2).

Equations and derivation.

1. **Observables and factorization** (3.1), pp. 3–4. With $(\boldsymbol{P}_x)_{i,j} = p_{j,x,i} = \mathbb{P}(X_1 = j, X_2 = x, X_3 = i)$ and $(\boldsymbol{P}_{x,x'})_{i,j} = p_{j,x,x',i}$,

$$\boldsymbol{P}_x = \boldsymbol{K}_x \boldsymbol{L}_x^\top, \qquad \boldsymbol{P}_{x,x'} = \boldsymbol{K}_{x'} \boldsymbol{D}_{x,x'} \boldsymbol{L}_x^\top,$$

$(\boldsymbol{K}_x)_{x',z} = k_z(x, x')$, $(\boldsymbol{L}_x)_{x',z} = \mu_z s_z(x') k_z(x', x)$, $\boldsymbol{D}_{x,x'} = \operatorname{diag}(k_1(x, x'), \dots, k_q(x, x'))$. *Our reconstruction:* $p_{j,x,i} = \sum_z \mu_z s_z(j) k_z(j, x) k_z(x, i)$ by total probability over Z; given $X_2 = x$ and Z, past and future are independent, and a fourth period inserts one diagonal factor $k_z(x, x')$. *Assumptions used:* first-order Markov given type; time homogeneity, so the same k_z appears in every transition.

2. **Assumption 1** (p. 3): "For each x the $r \times r$ matrix $\boldsymbol{P}_x$ has rank q." Equivalent to $\boldsymbol{K}_x$ and $\boldsymbol{L}_x$ both of column rank q; necessary $r \geq q$; the authors call it "directly testable from the data."

3. **Whitening** (3.2)–(3.3), p. 4. From the thin SVD $\boldsymbol{P}_x = \boldsymbol{U}_x \boldsymbol{E}_x \boldsymbol{V}_x^\top$, set $\boldsymbol{A}_x = \boldsymbol{E}_x^{-1/2} \boldsymbol{U}_x^\top$, $\boldsymbol{B}_x = \boldsymbol{E}_x^{-1/2} \boldsymbol{V}_x^\top$, so $\boldsymbol{A}_x \boldsymbol{P}_x \boldsymbol{B}_x^\top = \boldsymbol{I}_q$. With $\boldsymbol{Q}_x = \boldsymbol{A}_x \boldsymbol{K}_x$, $\boldsymbol{B}_x \boldsymbol{L}_x = \boldsymbol{Q}_x^{-\top}$ and

$$\boldsymbol{C}_{x,x'} \equiv \boldsymbol{A}_{x'} \boldsymbol{P}_{x,x'} \boldsymbol{B}_x^\top = \boldsymbol{Q}_{x'} \boldsymbol{D}_{x,x'} \boldsymbol{Q}_x^{-1}.$$

   If every $\boldsymbol{Q}_x$ were known in one common column order, (3.3) would give the kernel. *Assumption used:* Assumption 1.

4. **Each $\boldsymbol{Q}_x$ up to scale and order** (Step 2, pp. 4–5). Along a walk returning to x,

$$\boldsymbol{C}_{x_m,x} \cdots \boldsymbol{C}_{x_1,x_2} \boldsymbol{C}_{x,x_1} = \boldsymbol{Q}_x (\boldsymbol{D}_{x,x_1} \cdots \boldsymbol{D}_{x_m,x}) \boldsymbol{Q}_x^{-1},$$

   whose z-th eigenvalue is the probability of that return walk for type z, $k_z^m(x; x_1, \ldots, x_m; x)$, the product of its one-step kernel entries. *Assumption 2* (p. 3): for each x there is $\bar{m}$ such that the columns of $\boldsymbol{F}_x = (\boldsymbol{F}_{x,0}^\top, \ldots, \boldsymbol{F}_{x,\bar{m}}^\top)^\top$, stacking these probabilities by type, are all distinct. Then the joint diagonalizer is unique up to scale and order, $\tilde{\boldsymbol{Q}}_x = \boldsymbol{Q}_x \boldsymbol{\Omega}_x \boldsymbol{\Delta}_x$ (De Lathauwer et al., 2004, Thm. 6.1, as cited; not inspected by us). The scale follows from $\boldsymbol{B}_x \boldsymbol{p}_x = \boldsymbol{Q}_x^{-\top} \boldsymbol{\iota}_q$ and Lemma A.1 (p. 10).

5. **One common order** (Step 3, (3.4), p. 5). *Assumption 3* (p. 3): there is one x_0 such that every $x \neq x_0$ is connected to or from x_0 by a walk whose probability is non-zero for each z. Then $\tilde{\boldsymbol{Q}}_x^{-1} (\boldsymbol{C}_{x_m,x} \cdots \boldsymbol{C}_{x_0,x_1}) \tilde{\boldsymbol{Q}}_{x_0}$ is a permutation times an invertible diagonal matrix; its nonzero pattern reveals $\boldsymbol{H}_{x,x_0} = \boldsymbol{\Delta}_x^{-1} \boldsymbol{\Delta}_{x_0}$, and every state takes the order $\boldsymbol{\Delta}_{x_0}$.

6. **Theorem 1** (Step 4, p. 5). The diagonal of $\bar{\boldsymbol{Q}}_{x'}^{-1} \boldsymbol{C}_{x,x'} \bar{\boldsymbol{Q}}_x$ gives the kernel; with $\boldsymbol{p}_x^* = (p_{x,1}, \ldots, p_{x,r})^\top = \boldsymbol{K}_x \boldsymbol{\lambda}_x$ and $\boldsymbol{\lambda}_x = (s_1(x)\mu_1, \ldots, s_q(x)\mu_q)^\top$, the left inverse $\boldsymbol{K}_x^+$ (Assumption 1) gives $\boldsymbol{\lambda}_x$, hence μ_z and s_z. Under Assumptions 1–3, (μ_z, s_z, k_z) are identified from (X_1, X_2, X_3, X_4). Nothing requires $k_z(x, x') > 0$; the numerical example (p. 8) has absorbing states.

7. **Longer panels and higher order** (Remark and Corollary 1, p. 6; Sec. 6, p. 10). With T observations, histories are stacked in $\boldsymbol{K}_x$ and $\boldsymbol{L}_x$; at most $r^{(T-2)/2}$ types (T even) or $r^{(T-3)/2}$ (T odd); $q = \operatorname{rank} \boldsymbol{P}_x$ under Assumption 1. Order-p dependence needs $3 + p$ observations under "suitable modifications" of Assumptions 1–3, which are not stated.

8. **Binary illustrations** (5.5)–(5.6), pp. 7–10. With $\omega_z = k_z(1,2)$, $\varpi_z = k_z(2,2)$, $\pi_z = s_z(1)$: $\boldsymbol{P}_1$ is invertible iff $\omega_1 \neq \omega_2$ and $\pi_1(1-\omega_1)(1-\pi_2)(1-\varpi_2) \neq \pi_2(1-\omega_2)(1-\pi_1)(1-\varpi_1)$. The second part is a restriction on initial conditions (p. 7). With binary choice, a binary state and type-invariant state transitions (attributed to Magnac and Thesmar (2002), p. 9), $\boldsymbol{K}_x$ needs only one state where type CCPs differ; the $\boldsymbol{L}_x$ part is again an initial-conditions restriction (p. 9).

9. **Comparison** (Sec. 4, pp. 6–7). Kasahara and Shimotsu (2009) (Prop. 6 in published numbering) need six observations and, applied state by state, identify $\boldsymbol{K}_x \boldsymbol{\Delta}_x$ with a different unknown permutation at each x, which does not recover kernels, initial distributions or type probabilities (p. 6). Hu and Shum (2012) need $r = q$ and monotonicity restrictions, which HJ do not (pp. 6–7). *Check:* this likely corresponds to working-paper Proposition 5; the critique targets the per-state route, not the baseline Proposition 1, which uses one global eigendecomposition.

10. **Estimator** (Appendix B, p. 11): maximum likelihood by EM on $\sum_z \mu_z \ell_z$, with type likelihood $\ell_z = \prod_x s_z(x)^{\{x_1 = x\}} \prod_{x,x'} k_z(x,x')^{n_{x,x'}}$ and transition counts $n_{x,x'}$; closed-form M-step for μ_z,

s_z, k_z ($r + r^2$ parameters per type), multiple starting values, outer-product-of-score standard errors. *Note:* p. 11 calls s_z "the steady-state distribution"; we read it as the initial distribution of Section 2.

Results to note. One Monte Carlo design (Tables B.1–B.2, p. 13): $r = q = 2$, $\mu_1 = 0.4$, transition probabilities 0.2–0.8, steady-state initial draws, $n = 500$, 10,000 replications. At $T = 4$, s.d. of $\hat{\mu}_1$ is 0.068 (mean 0.414); at $T = 5$, 0.046; t-test sizes 0.025–0.053. Here $|\omega_1 - \omega_2| = 0.6$, so Assumption 1 holds by a wide margin; there is no rare-event, large-r or weak-rank design. *Check:* the tables label two columns $k_2(0, 1)$, $k_2(1, 1)$, which the true values identify as $k_2(1, 2)$, $k_2(2, 2)$.

Assumptions: stated and implicit.

- *Stated by the authors:* finite, known q and $\mu_z > 0$; first-order, time-homogeneous Markov dependence given type; free initial distributions; Assumptions 1–3; random sample of complete sequences of common length (p. 11).
- *Implicit, in our reading:* "testable" means testable in the population; no rank test or near-singular analysis is given; Assumptions 1–2 must hold at every state used, which is demanding with large r and sparse cells; long walks multiply many estimated matrices; no result for time-varying kernels; no attrition; recovering utilities still needs β, the shock distribution and a reference-alternative normalization per type (Magnac and Thesmar, 2002).

Question for discussion. With deterministic vehicle age, only keepers one month younger lead into a state, so the $\boldsymbol{L}_x$ half of Assumption 1 becomes a condition on how the types arrived at each age, through other state variables or type-specific responses to them (our reconstruction); with rare purchases, is that variation detectable in samples, or would the type split rest on parametric structure?

A.2.7 Aguirregabiria et al. (2021): sufficient statistics for fixed effects in dynamic logit

Why it is included: it states which payoff parameters of a forward-looking keep-or-replace logit remain identified when persistent heterogeneity in payoffs and in the discount factor, and its dependence on the initial state, are left completely unrestricted, which shows what a random-effects type mixture identifies only through its distributional assumption.

Setup. Theoretical econometrics with a Monte Carlo and a re-estimation of Rust's bus data; $N \to \infty$, T fixed, infinite horizon (pp. 280–285). Choice $y_{it} \in \mathcal{Y} = \{0, 1, \ldots, J\}$; objective $\mathbb{E}_t[\sum_j \delta_i^j \Pi_{i,t+j}(y_{i,t+j})]$ (p. 282). *Given state:* endogenous $x_{it} = (y_{i,t-1}, d_{it})$ (lagged choice and duration in the current choice) and exogenous Markov z_{it}, observed by agent and researcher. *Unobserved by the researcher:* heterogeneity $\theta_i = (\eta_i, \delta_i)$ and shocks $\varepsilon_{it}(j)$. *Notation:* the discount factor is δ_i; β denotes payoff functions of the endogenous state. In machine replacement, keep is $y = 1$ (carries duration) and replace is $y = 0$. Method: Andersen/Chamberlain conditional maximum likelihood on a sufficient statistic for θ (FE-CML).

Equations and derivation.

1. **Payoff** (1)–(2), pp. 282–283: $\Pi_{it}(j) = \alpha(j, \eta_i, z_{it}) + \beta(j, x_{it}, z_{it}) + \varepsilon_{it}(j)$, with

$$\beta(j, x_{it}, z_{it}) = 1\{j = y_{i,t-1}\}\beta_d(j, d_{it}, z_{it}) + 1\{j \neq y_{i,t-1}\}\beta_y(j, y_{i,t-1}, z_{it}), \qquad \beta_y(j, j, z) = 0.$$

Assumptions used: Asm. 1(B), η additively separable from x, α unrestricted, called "key" (p. 283); Asm. 1(C), $\beta_d(0, d, z) = 0$, no duration dependence in alternative 0; additive switching plus duration, "not without loss of generality" (p. 283).

2. **Transition** (3), p. 283: $d_{i,t+1} = 1\{y_{it} = y_{i,t-1}\}d_{it} + 1$ if $y_{it} > 0$, and 0 if $y_{it} = 0$. *Assumptions used:* deterministic (App. B, p. 309, adds i.i.d. jumps $\xi^d \geq 0$); Asm. 1(E), z exogenous, time-homogeneous Markov, and constant over consecutive periods (p. 285).

3. **Choice probability** (5)–(6), pp. 284–285. Since d_{t+1} is a function of (j, x_{it}),

$$P_{\theta_i}(j \mid x_{it}, z_{it}) = \frac{\exp\{\alpha(j, \eta_i, z_{it}) + \beta(j, x_{it}, z_{it}) + v_{\theta_i}(j, d_{t+1}[j, x_{it}], z_{it})\}}{\sum_{k \in \mathcal{Y}} \exp\{\alpha(k, \eta_i, z_{it}) + \beta(k, x_{it}, z_{it}) + v_{\theta_i}(k, d_{t+1}[k, x_{it}], z_{it})\}}.$$

   Assumptions used: Asm. 1(A), infinite horizon, $\delta_i \in (0,1)$ heterogeneous; Asm. 1(D), EV1 i.i.d. over (i,t,j). Property 1 (p. 285): with $\beta_d = 0$, $v_\theta(j,z)$ does not depend on y_{t-1}. Property 2: under Asm. 2 (step 7), $v_\theta(j, d_{t+1}, z) = v_\theta(j, d_j^*, z)$ for $j = y_{t-1}$, $d_t \geq d_j^* - 1$.

4. **Binary reparametrization** (Lemma 1, (7); (11), p. 287): $y_t = 1\{\tilde{\alpha}_\theta + y_{t-1}\tilde{\beta}_d(d_t) + \tilde{v}_\theta(d_t + 1) + \tilde{\varepsilon}_t \geq 0\}$, with $\tilde{\beta}_d(d) = \beta_d(1,d) - \beta_y(1,0) - \beta_y(0,1)$ and $\tilde{v}_\theta(d) = v_\theta(1,d) - v_\theta(0,0)$. Marginal returns $\beta_d(d) - \beta_d(d-1) = \tilde{\beta}_d(d) - \tilde{\beta}_d(d-1)$, $d \geq 2$.

5. **Sufficient-statistic likelihood** (8)–(10), p. 286. For a history $\tilde{y} = (d_1, y_0; y_1, \ldots, y_T)$, if $\ln P(\tilde{y} \mid \theta, \beta) = U(\tilde{y})'g_\theta + S(\tilde{y})'\beta^*$, then

$$\ln P(\tilde{y} \mid \theta, \beta, U) = S'\beta^* - \ln\Big(\textstyle\sum_{\tilde{y}': U(\tilde{y}') = U} \exp\{S(\tilde{y}')'\beta^*\}\Big),$$

   free of θ. Lemma 2 (p. 287): β^* is identified iff there are $K+1$ histories with the same U and a nonsingular matrix of S-differences. *Assumptions used:* Asm. 1(F), $\theta \mid (z\text{-path}, x_{i1})$ unrestricted, so $p(y_0, d_1 \mid \theta)$ is free (p. 286); EV1 for the exponential-family form. *Note:* as printed, (8) uses $\sum_{t=1}^{T}$; a history probability is a product, and (9) treats it as one.

6. **Forward-looking binary model** (20), p. 289, with $\sigma_\theta(y_{t-1}, d_t) \equiv -\ln(1 + \exp\{\tilde{\alpha}_\theta + y_{t-1}\tilde{\beta}_d(d_t) + \tilde{v}_\theta(d_t + 1)\})$:

$$\ln P(\tilde{y} \mid \theta) = \ln p_\theta(y_0, d_1) + \sum_{t=1}^{T} y_t\big[\tilde{\alpha}_\theta + y_{t-1}\tilde{\beta}_d(d_t) + \tilde{v}_\theta(d_t + 1)\big] + \sigma_\theta(y_{t-1}, d_t).$$

   Proposition 4 ((21), p. 290; proof (A.15), p. 305): $U = [d_1, y_0, \{H^{(1)}(d), \Delta^{(1)}(d) : d \geq 1\}]$, with $H^{(1)}(d) = \sum_t 1\{y_{t-1} = 1, d_t = d\}$ and $\Delta^{(1)}(d) = 1\{y_T = 1, d_{T+1} = d\} - 1\{y_0 = 1, d_1 = d\}$ (Table 1, p. 287). These are also the only statistics multiplying $\tilde{\beta}_d$, so *no* structural parameter is identified. The term $\tilde{v}_\theta(d_t + 1)$ puts θ at every duration, and $v_\theta(1,n)$ is nonlinear in θ (fn. 19).

7. **Assumption 2 and Proposition 5** ((24), p. 290; (A.16)–(A.18), p. 305). Asm. 2 (p. 284): $\beta_d(j,d) = \beta_d(j, d_j^*)$ for $d \geq d_j^*$. By Property 2 the θ-functions are constant for $d \geq d^*$, so only sums enter U:

$$U = \Big[d_1, y_0, \{H^{(1)}(d), \Delta^{(1)}(d) : d \leq d^* - 1\}, \textstyle\sum_{d \geq d^*} H^{(1)}(d), \sum_{d \geq d^*} \Delta^{(1)}(d)\Big],$$
$$S = H^{(1)}(d^*) + \Delta^{(1)}(d^*), \qquad \beta^* = \beta_d(1, d^* - 1) - \beta_d(1, d^*).$$

   By Lemma 3(v), $S = X^{(1)}(d^* - 1)$, the number of keeps at duration $d^* - 1$ (our reading). *Note:* signs vary (Prop. 5, Tables 7–8); identification is unaffected. Footnote 19: "it seems" the only restrictions making $\tilde{\beta}_d(n) - \tilde{\beta}_d(n-1) + \tilde{v}_\theta(n+1) - \tilde{v}_\theta(n)$ in Example 4(b), (23), free of θ are those making it zero, a conjecture, not a theorem.

8. **Identifying** d^* (Prop. 6, (26)–(27), p. 291): for $A_n = \{0,0 \mid 1_{n-1}, 0, 1_{n+1}\}$, $B_n = \{0,0 \mid 1_n, 0, 1_n\}$, $2n + 1 \leq T$, $d^* = \max\{n : \ln P(A_n) - \ln P(B_n) \neq 0\}$, provided $d^* \leq (T-1)/2$.

9. **Estimation** (43)–(48), pp. 297–298: the conditional log-likelihood $\mathcal{L}_N(\beta^*)$ is globally concave, $\sqrt{N}(\hat{\beta}^* - \beta^*) \Rightarrow N(0, J^{-1})$; d^* is chosen by $\text{BIC}_N(d^*) = \ell_N(d^*) - \frac{d^*}{2} \ln N$ (consistent, Prop. 13). Enumeration cost is exponential in T (p. 297).

10. **Distribution of** θ (Sec. 3.7, (40)–(42), p. 296): payoff levels and marginal effects need a finite-support restriction or set identification. *Check:* the paper says $\tilde{v}_\theta$ can then be solved from the Bellman equation, but that requires the whole β_d function and δ_θ, which the CMLE does not identify; this step is unresolved.

Results to note. Tables 2–3 (pp. 291, 295): without duration dependence, switching costs are identified with $T \geq 3$, myopic or forward-looking (Props. 1, 2, 7, 8); myopic with duration dependence, all $\tilde{\beta}_d(d)$, $1 \leq d \leq T-2$ (Props. 3, 9); forward-looking with duration dependence, nothing in the binary model (Prop. 4); under Asm. 2, one increment per alternative (Props. 5, 11). Monte Carlo (Table 5, p. 300; $RC_i \sim N(8, 2^2)$, $\delta = 0.95$, $d^* = 3$, $N = 1000$, truth 1): CMLE 1.007/0.999/0.995 (s.d. 0.144/0.080/0.073) in samples A/B/C; a two-type MLE ignoring initial conditions 0.978/0.896/0.857; Hausman rejection at 5% 0.042/0.308/0.515, so no power with one spell per bus. Rust data (Tables 6, 8, 9): 46 of 104 buses contribute; RC_i is absorbed by $\tilde{\alpha}_\theta$ and not estimated; $d^* = 3$, $\hat{\beta}_{d^*} = 1.7009$ (1.0244) against MLE 0.36–0.45; Hausman $p = 0.19$–0.22.

Assumptions: stated and implicit.

- *Stated by the authors:* Asm. 1(A)–(F); Asm. 2 for forward-looking duration effects; stationarity, which "can be unrealistic" (p. 283); z constant within the window (p. 285); T fixed, $N \to \infty$, bounded parameter spaces (p. 297).

- *Implicit, in our reading:* flattening serves as an identifying device, and since BIC selects $d^* \leq (T-1)/2$, a short panel can only report a small d^*; the effective sample is the contributing units, which in our toy enumeration need two observed renewals (shortest $T = 2d^* + 1$ from a fresh start; pattern checked for $d^* \leq 5$, not proved); Monte Carlo replacement rates and contributing shares are not reported; Sec. 3.7 omits the unidentified parts of β_d and δ.

Question for discussion. In a monthly household keep/replace model whose replace payoff does not depend on vehicle age, the only flattening point is the state cap on age: is there an economic reason for a flat age cost after a few years, or would FE-CML identification there be an artifact of truncation?

A.3 C3. Latent household state

A.3.1 Li et al. (2011): latent financial states and bank cross-selling

We work from the authors' November 2010 manuscript (third revision), not the published article, so we cite equations, tables, and sections; page numbers are manuscript pages, and "A" marks appendix equations. *Why it is included:* it is a worked template, in bank data, for a latent time-varying household state with a covariate-dependent transition equation and a state-dependent measurement equation for monthly product openings.

Setup. One bank, 2,000 calibration households, monthly, November 2003 to January 2006 (27 months), seven product categories (Sec. 3). *Given state:* a latent "financial state" $s \in \{1, \ldots, S\}$, occupied by the household but observed by neither the bank nor the researcher; observed covariates (balance changes, accounts, transactions, tenure, income). *Actions:* household openings Y_{ijt}; bank solicitations Z_{ijkt} for product j by channel k (mail, email), observed by the researcher. *Objectives:*

the household maximizes current utility only; the bank maximizes discounted profit (Sec. 5). The state is measured only through openings.

Equations and derivation.

1. **Latent utility** (1)–(2). $Y_{ijt} = 1$ if i opens j in month t, with

$$\begin{aligned} U_{ijt}(s) = {} & \beta_{0ij}(s) + \textstyle\sum_k \big(\beta_{1ij}(s) + \beta_{2ik}(s)\big) Z_{ijkt} + \beta_{3i}(s) \sum_k \sum_j \sum_{\tau=1}^{t-1} Z_{ijk\tau} \\ & + \beta_{4i}(s)\mu_{\Delta BAL_{i(t-1)}} + \beta_{5i}(s)\sigma_{\Delta BAL_{i(t-1)}} + \beta_{6i}(s) NACCT_{ijt} + \beta_{7i}(s) TRANS_{i(t-1)} \\ & + \cdots + \beta_{10i}(s) INCOME_{it} + \varepsilon_{ijt}(s), \end{aligned}$$

   where the omitted terms are $TENURE_{it}$ and $COMP_{it}$. *Assumptions used:* every coefficient is state-specific; lagged balance-change and transaction covariates are exogenous to solicitations (fn. 3); a solicitation shifts current utility and the cumulative stock shifts later utility. *Note:* (2) has no continuation value, so the household is myopic given its state.

2. **Orthant probability** (3)–(4). $\boldsymbol{\varepsilon}_{it}(s) \sim MVN[0, \Sigma]$ across the J products, and $\mathbb{P}(\mathbf{Y}_{it} \mid s) = \int_{M_j} \phi_J(\boldsymbol{\varepsilon}_{it}(s); 0, \Sigma)\, d\boldsymbol{\varepsilon}_{it}(s)$, with $M_j = (-\infty, 0)$ if $Y_{ijt} = 0$ and $(0, \infty)$ otherwise, writing ϕ_J for the printed normal density. *Assumptions used:* $Y_{ijt} = \mathbf{1}\{U_{ijt}(s) > 0\}$; errors independent over time given s; Σ has unit diagonal (App. Sec. 2). *Check:* with $U = \bar{U} + \varepsilon$, $Y_{ijt} = 1$ iff $\varepsilon_{ijt} > -\bar{U}_{ijt}(s)$, so the printed M_j give the sign of U and must be shifted by $-\bar{U}_{ijt}(s)$ to apply to ε. For $J = 1$, (4) is $\Phi(\bar{U}_{i1t}(s))$ or its complement; unconditionally, $\mathbb{P}(\mathbf{Y}_{it}) = \sum_s \mathbb{P}_{it}(s)\mathbb{P}(\mathbf{Y}_{it} \mid s)$.

3. **Transition matrix** (5). M_{it} is $S \times S$ with zero diagonal; P_{itmn} is the probability of a jump from m to n. *Assumptions used:* first-order switching among states common to all households; persistence comes from waiting times. *Note:* a discrete-time version with self-transitions gave "very similar results" (fn. 4), not shown.

4. **Gamma waiting times** (6)–(7). $W_{it}(s) \sim$ Gamma with shape $\lambda_{it}(s)$ and inverse scale $k_i(s)$, so $E[W_{it}(s)] = \lambda_{it}(s)/k_i(s)$. *Check (hazard):* $h(w) = f(w)/\{1 - F(w)\}$. For $\lambda = 1$, $f = ke^{-kw}$ and $1 - F = e^{-kw}$, so $h = k$, with no duration dependence; for $\lambda > 1$ the hazard rises with w, for $\lambda < 1$ it falls, both toward k. A monthly Markov chain with self-transitions forces geometric durations, so duration dependence requires the semi-Markov form (state plus elapsed time).

5. **Covariate-dependent shape** (8)–(9). $\log \lambda_{it}(s) \sim N(\bar{\lambda}_{it}(s), \sigma^2_\lambda)$, with $\bar{\lambda}_{it}(s)$ linear in experience $TACCT_{i(t-1)}$, current solicitations ($\alpha_{2ij}(s)$), the cumulative channel stock and its square, $COMP$, $INCOME$, and closed accounts. *Assumptions used:* only the shape moves; $k_i(s)$ does not. $\alpha_{2ij}(s) < 0$ shortens the stay in s, the "educational" channel the authors call speculative (p. 13). Past purchases enter through $TACCT$, so the state depends on the household's own choices. *Note:* the text resets the stock at entry to the current state; the printed sum starts at $t_s = 1$.

6. **Priors and initial state** (10)–(12). Transition rows and initial probabilities are Dirichlet with log-normal hyperparameters, and $\bar{\eta}_{is} = \omega_{0i} + \omega_{1i} TACCT_{i0} + \omega_{2i} ASSET_{i0}$. *Note:* (12) is a parametric initial-condition equation in start-of-sample holdings; elapsed time in the first spell (left censoring) is not treated.

7. **Hierarchy** (13). $\theta_{im} = \mu_{m0} + \mu_{m1} AGE_i + \mu_{m2} GENDER_i + \mu_{m3} HSIZE_i + e_{im}$, $e_i \sim N[0, \Omega]$, for the stacked utility parameters. The heterogeneity distribution of $\alpha_i(s)$, $k_i(s)$, ω_i is not specified.

8. **Endogenous solicitations** (A1)–(A2). $Z_{ijkt} \mid \gamma_i \sim$ Poisson(γ_i), with $\ln \gamma_i$ linear in state-averaged coefficients on $NACCT$, $TENURE$, $INCOME$ and averaged hierarchy coefficients.

Assumptions used: the bank knows average response parameters but not the current state, so Z is independent of s and ε given Θ_i. *Check:* the last three terms of (A2) average population hyperparameters and cannot be separated from φ_0; (A1) treats Z as a count, elsewhere it is binary.

9. **Likelihood** (A3)–(A4). For the realized latent path D_{it},

$$\mathbb{P}(\mathbf{Y},\mathbf{Z}) = \prod_i \prod_s (\pi_i(s))^{\mathbf{1}\{D_{i0}=s\}} \Big\{ \prod_t \mathbb{P}(Y_{it} \mid D_{it}, Z_{it}) \mathbb{P}(Z_{it}) \cdot \Big(\prod_l P_{itsl}^{\mathbf{1}\{D_{it}=s \,\&\, D_{i(t+1)}=l\}} \Big) \prod_k f(W_{itk} \mid D_{it}, \lambda_{it}, \kappa_i) \Big\},$$

where (A4) is a gamma density restricted to the interval between jump times τ_{k+m} and τ_{k+m+1}. *Assumptions used:* complete-data likelihood with MCMC data augmentation. *Note:* samplers, burn-in, and convergence checks are not reported; (A4) writes the rate κ_i, (6) writes $k_i(s)$.

10. **Ordering and number of states** (A5), Table 3A. With mean-centered covariates,

$$\tfrac{1}{J}\textstyle\sum_j \beta_{0ij}(1) \le \tfrac{1}{J}\sum_j \beta_{0ij}(2) \le \cdots \le \tfrac{1}{J}\sum_j \beta_{0ij}(S).$$

This fixes labels, not S or the transition law. Log marginal densities for $S = 1,\ldots,4$ are $-1{,}154{,}047.7$, $-1{,}144{,}416.0$, $-984{,}746.9$, $-1{,}184{,}113.2$, so $S = 3$. *Check:* the sequence is not monotone, and its computation is not described.

11. **Bank dynamic program** (15)–(18). Expected profit (15) weights state-specific purchase probabilities by filtered $\mathbb{P}_{it}(s)$, margins r_{ij}, less channel costs c_k; (16) sums it over the 27 months; (17) adds an extreme-value shock, giving the logit policy (18). *Note:* the DP is solved with estimated customer parameters, not estimated from bank behavior. As printed, $(1+\delta)^{t-\xi_1}$ in (16) compounds rather than discounts, and δ is not reported.

Results to note. $S = 3$ fits best (Table 3A); the full model has the best validation purchase hit rate, 0.268 against 0.101–0.246 (Table 3B). Mean balance change raises utility (1.06 to 1.29) and its dispersion lowers it (-1.45 to -1.92) (Table 4A). Mean waiting times are 9.68, 10.90, 15.07 (Table 4D; units presumably months), so 27 months cover roughly two or three spells, the first left-censored. The simulated dynamic policy raises profit from 22,021.5 to 60,925.5 (Table 6) under the same fitted model.

Assumptions: stated and implicit.

- *Stated by the authors:* S common states; zero-diagonal semi-Markov switching with gamma durations; MVN errors, unit-diagonal Σ; solicitations exogenous given average parameters (A1)–(A2); balances exogenous (fn. 3); ordering (A5); finite-horizon bank problem.
- *Implicit, in our reading:* myopic customers; states identified from rare openings (about 3.1% of observations, p. 19), with persistent heterogeneity (13) and time-varying states competing for the same serial correlation (Heckman, 1981); balances shift utility rather than signal the state; no bank targeting on the current state.
- *Reporting problems:* Table 2 prints a mean balance change (-6.35) outside its range (-0.033 to 3.8), $TACCT$ up to 155 with seven categories, and tenure in years with mean 66.46. Table 4A's waiting-time intercepts (-8.23, -8.53, 13.43) are not consistent log shapes for Table 4D; combining Table 4D with Table 4A's $k_i(s)$ through (7) gives shapes 61.66, 54.94, 16.88 (our computation), nearly deterministic durations.

Question about the paper.

- With openings as the only measurement, what separates the latent financial state from the persistent heterogeneity in (13) over 27 months?

A.4 C4. Liquidity and purchase

A.4.1 Adams et al. (2009): liquidity constraints in subprime auto purchases

Why it is included: it is direct evidence that the up-front cash outlay, separately from the financed price, governs purchase: subprime used-car applicants respond strongly to the required down payment and to tax-refund season, and barely to the financed price.

Setup. Unit: a loan application at one large subprime dealer-lender, June 2001 to December 2004 (p. 52); over 50,000 applications, 34 percent purchase (Table 1). *Outcome:* $q_i = 1$ if applicant i buys, given that i applied. *Set by the company:* minimum down payment d_i (by credit category and region) and list price l_i (a markup over cost). *Negotiated:* price p_i. *Chosen by the buyer:* buy or not, and $D \geq d$. Assets and debt are not observed (p. 53).

Equations and derivation.

1. **Purchase probit** (1), p. 59: $q_i = 1 \Leftrightarrow q_i^* = \mathbf{x}_i'\boldsymbol{\alpha} + \varepsilon_i \geq 0$, where $\mathbf{x}_i$ holds offer terms (price, rate, term, minimum down), car and applicant characteristics, and city, credit-category, month and year dummies. *Assumptions used:* ε_i i.i.d. normal; one binary decision on the lot; terms affect neither who applies (they are not posted) nor car choice (p. 60).

2. **Missing prices** (p. 59, fn. 17). *Assumption used:* a nonpurchaser faced the car and price of a random purchaser in the same city, week, credit and income category. *Note:* prices are imputed for about two thirds of the sample, which attenuates the price effect; fn. 17 notes a further bias toward underestimating it.

3. **Minimum-down identification** (p. 60). Given the dummies, the effect comes from company changes over time and differential changes across categories and cities. *Assumption used:* company decisions are uncorrelated with ε_i given controls, since analysts see what headquarters sees. *Implicit, in our reading:* minimums are not timed to demand shifts the month dummies miss (they are about \$150 higher in February, p. 57).

4. **Negotiated-price equation** (2), p. 60: $p_i = l_i\lambda_l + \mathbf{x}_i'\boldsymbol{\lambda}_x + \nu_i$, estimated jointly with (1) by maximum likelihood, (ε_i, ν_i) correlated normal. *Assumptions used:* l_i is excluded from (1) given cost controls; it varies through changes in the margin schedule, regions, and discontinuities of the margin formula in cost. Relevance is strong ($t > 100$); exclusion is not testable. Columns 3–4 use cost-bucket and state dummies instead (fn. 18).

5. **Expected payment without a liquidity constraint** (3), p. 61:

$$\mathbb{E}[Payment] = D + (p - D)\,\frac{\sum_{t=1}^{T}(1+s)^{-t}S_t}{\sum_{t=1}^{T}(1+r)^{-t}} = D + (p - D)\phi,$$

with s the buyer's monthly discount rate, S_t the probability of no default before the end of month t, and ϕ the expected present value paid per dollar borrowed; the denominator is the annuity factor (fn. 7). *Assumption used:* an unconstrained buyer at the minimum ranks offers by $\mathbb{E}[Payment]$. *Reconstructed step:* ΔD changes it by $\Delta D(1-\phi)$ and Δp by $\Delta p\,\phi$, so indifference requires $\phi = \Delta D/(\Delta D + \Delta p)$. *Check:* \$100 of down payment worth about \$30 of price at 5 percent (p. 61) implies $\phi \approx 0.77$.

6. **Coefficients as a liquidity test** (p. 63). The ratio of the minimum-down to the price effect is the empirical Δp per \$100 of ΔD: \$2,884, so $\phi \approx 0.034$, which (3) matches only at a 1,415 percent annual discount rate (474 percent with col. 4); at 10, 20 and 50 percent the benchmarks are \$38, \$55, \$108. *Note:* the col. 2 price effect, −0.0010 (0.0011), is not distinguishable from zero, so the ratio is bounded below rather than estimated, and neither \$2,884 nor 1,415 percent has a reported standard error. The rounded coefficients give about \$2,990. S_t and the compounding convention are not reported (fn. 19), so we read 1,415 percent as an order of magnitude.
7. **Cell-level demand** (4), p. 64: $\ln Q_j = \beta_A \ln A_j + \tilde{\mathbf{x}}_j'\boldsymbol{\beta}_x + u_j$, cells = city × week × credit category × income range; OLS (col. 5) and IV with average list price (col. 6). It avoids the imputation of step 2; dividing by three roughly converts to the probit scale (Table 2, note d).
8. **February evidence** (pp. 56–59). Applications are 23 percent higher and the close rate is 40 versus 33 percent, almost 50 percent more sales (p. 50). Across 12 income-by-dependents groups the February increase rises with the group's earned income tax credit (Fig. 2B); trade-in purchases show no seasonality. *Assumption used:* the spike reflects tax refunds. *Note:* a descriptive gradient without standard errors; whether February buyers would otherwise buy later is not observed.

The default, loan-size and adverse-selection analyses, (5)–(9), are outside our current scope.

Results to note (Table 2, col. 2, dF/dx, per \$100 where applicable). Minimum down −0.0299 (0.0006), about 9 percent of the 0.34 purchase rate; price −0.0010 (0.0011); February 0.1594 (0.0047) relative to December. Robustness subsamples give −0.0238 to −0.0304 (p. 65); cell level −0.0895 (0.0039) log points (col. 5). Standard errors use 50 bootstrap replications. *Reconciliation:* the text puts February demand 30 percent higher (p. 59), but 0.1594/0.34 is larger; text and Fig. 2B differ on the top income group (\$3,500 versus \$3,000).

Assumptions: stated and implicit.

- *Stated by the authors:* normal errors; one binary decision; minimums exogenous given dummies; list price a valid instrument; imputed nonpurchaser prices; no effect of terms on applications or car choice.
- *Implicit, in our reading:* purchase is one-shot, not a choice of when to buy; impatience or present bias is set aside by plausibility, not a test; the dealer's applicants represent the population of interest.

Question about the paper.

- Given the imprecise price effect, do we cite \$2,884 as a lower bound, or only the down-payment effect?

A.5 C5. Initial conditions and baseline

A.5.1 Heckman (1981): initial conditions and incidental parameters

Why it is included: it supplies the likelihood term needed when units enter the data partway through their histories and differ in a persistent way the analyst does not observe: a model for the first observed state given the household's type.

Setup. Individuals $i = 1, \ldots, I$ follow a binary first-order Markov process over $t = 1, \ldots, T$ that starts at a fixed $d(i, 0)$, but the analyst observes it only from period $J > 0$ onward (pp. 179, 183).

Observed: $d(i,J),\ldots,d(i,T)$, and in extensions covariates $\mathbf{Z}(i,t)$. *Unobserved effect:* a persistent $\tau(i)$ with density $f(\tau)$. No decision problem is specified.

Equations and derivation.

1. **Latent index model** (4.1), p. 181:

$$Y(i,t)=\beta_0+\gamma d(i,t-1)+\tau(i)+U(i,t),\qquad d(i,t)=1 \text{ iff } Y(i,t)\geq 0.$$

   γ is state dependence; $\tau(i)$ makes the composite error serially correlated. *Assumptions used:* $E[U]=E[\tau]=0$, $\sigma_U^2=1$, $E[\tau^2]=\sigma_\tau^2$; U serially independent and independent of τ; $d(i,0)$ fixed and nonstochastic (p. 181).

2. **Transition given** τ (4.2), p. 181:

$$\mathbb{P}[d(i,t)\mid d(i,t-1),\tau(i)]=\Phi\{[\beta_0+\gamma d(i,t-1)+\tau(i)][2d(i,t)-1]\}.$$

   Assumptions used: U normal, which the author calls "inessential" (p. 181).

3. **Probability of the first observed state.** Equation (4.3), p. 182, sums the probabilities of all paths from $d(i,0)$ to $d(i,J)$, so $P[d(i,J)\mid\tau]$ depends on β_0, γ, $d(i,0)$ and τ (p. 184). In the stationary case, (4.4), p. 182,

$$\Pi_1(\tau)=\frac{\Phi(\beta_0+\tau)}{1-\Phi(\beta_0+\gamma+\tau)+\Phi(\beta_0+\tau)},\qquad \Pi_0=1-\Pi_1.$$

   Assumptions used: bounded β_0,γ,τ (ergodicity) and equilibrium by period J. *Check* (from fn. 1): stationarity $\Pi_1=p_{11}\Pi_1+p_{01}(1-\Pi_1)$ with $p_{01}=\Phi(\beta_0+\tau)$, $p_{11}=\Phi(\beta_0+\gamma+\tau)$ gives (4.4), which is not a probit (p. 182). *Note:* (4.3) is printed as a product of sums; we read it as nested sums over paths.

4. **Correct likelihood** (4.7), p. 183, for a sample starting at J:

$$\mathcal{L}=\prod_{i=1}^{I}\int_{-\infty}^{\infty}\prod_{t=J+1}^{T}\Phi\{[\beta_0+\gamma d(i,t-1)+\tau][2d(i,t)-1]\}\,P[d(i,J)\mid\tau]\,f(\tau)\,d\tau.$$

   The conditional version (4.6) divides by $\int P[d(i,J)\mid\tau]f(\tau)\,d\tau$. Either is consistent as $I\to\infty$ for $T\geq 2$ (p. 184). *Assumptions used:* parametric f (normal "for convenience," p. 182).

5. **Naive likelihood** (4.8), p. 184: the same expression with $P[d(i,J)\mid\tau]$ omitted, treating $d(i,J)$ as exogenous. It is inconsistent for $\beta_0,\gamma,\sigma_\tau^2$ (p. 184); the proof is only sketched (fn. 3). *Check* (our reconstruction): by Bayes' rule $P[d(i,J)\mid\tau]f(\tau)\propto f(\tau\mid d(i,J))$, so (4.6) integrates over τ *given* the first state and (4.8) over $f(\tau)$. They agree only if $P[d(i,J)\mid\tau]$ is flat in τ: degenerate f, or a new process at J (p. 179). Those starting in state 1 have higher τ, and (4.8) credits that persistence to γ.

6. **Exact solutions and their data requirements** (pp. 183–185). (a) The full history from $d(i,0)$, likelihood (4.5), p. 182, is "hard to come by" (p. 184). (b) Initial stationarity, (4.4) in (4.7), is "computationally forbidding" (p. 184) and, with $\mathbf{Z}(i,t)\boldsymbol{\beta}$ in place of β_0, needs a stationary $\mathbf{Z}$ process, which "excludes time and age trends" (p. 184). (c) Presample $\mathbf{Z}$ can be integrated over a finite-parameter density g estimated elsewhere; plugging in means is not correct, because $\mathbf{Z}$ enters nonlinearly (p. 185).

7. **Fixed effect likelihood** (4.9), p. 186. Treating each $\tau(i)$ as a parameter, $P[d(i,J) \mid \tau(i)]$ cancels from (4.6):

$$\mathcal{L} = \prod_{i=1}^{I} \prod_{t=J+1}^{T} \Phi\{[\beta_0 + \gamma d(i,t-1) + \tau(i)][2d(i,t) - 1]\}.$$

   It needs no presample data and no f, but stayers give $\hat{\tau}(i) = \pm\infty$ and drop out, and for fixed T the inconsistent $\hat{\tau}(i)$ contaminate $\hat{\gamma}, \hat{\beta}$: the incidental-parameters problem (pp. 186–187). *Check* (fn. 9, p. 187): with $\gamma = 0$, $T = 2$ and a period effect β, movers give $\hat{\tau}(i) = -\beta/2$ and, as $\sigma_\tau^2 \to 0$, $\operatorname{plim} \hat{\beta} = 2\beta$.

8. **Approximate solution** (pp. 188–189). Model the first state by a reduced-form probit, $Y(i,J) = \sum_{l=0}^{J} D(l)[\mathbf{Z}(i,l)] + \mu(i,J)$, $D(l)$ polynomials in practice (fn. 11), with $\mu(i,J)$ freely correlated with $\tau(i) + U(i,t)$, $t > J$, and no cross-restrictions to the structural parameters. *Assumptions used:* consistency for $I \to \infty$, fixed T, holds "assuming that the model is exact" (p. 189). *Note:* by step 3 the true marginal is not a probit, so in our reading consistency is not guaranteed; the correlation's parameterization is not stated.

Results to note. Monte Carlo, 25 replications of $I = 100$, no dispersion reported (pp. 189–193). Without a lagged state (Table 4.1, p. 191), the fixed effect probit at $T = 8$ is biased at most 10%, toward zero; the third design reads $\beta = -0.1$ in the table header but $\beta = 0.1$ in the text (p. 189), and the entries $(-0.10, -0.09, -0.10)$ fit -0.1. With a lagged state (Table 4.2, p. 193), fixed effects at $T = 8$ give $\hat{\gamma}$ of 0.03 to 0.25 for true 0.5 and the wrong sign in all six cells for true 0.1; the approximate estimator at $T = 3$ gives 0.54 to 0.70 for 0.5. All estimators are biased at $I = 100$, $T = 3$ (p. 194). *Illustration with invented parameters, not evidence* (seed 20261023, $I = 1000$, 100 replications, true $\gamma = 0.5$; mean $\hat{\gamma}$, standard deviation in parentheses): with 3 observed periods, an exogenous first state gives 1.350 (0.077), the stationary correction 0.507 (0.172), the approximate initial probit 0.503 (0.195); with 8 periods, 0.632 (0.056) versus 0.502 (0.049) for both corrections. With true $\gamma = 0$ and 8 periods, a pooled probit gives 0.880 (0.046).

Assumptions: stated and implicit.

- *Stated by the author:* first-order Markov binary state; $\tau + U$ components of variance with U serially independent and independent of τ; normal U and, for convenience, normal f; bounded exogenous $\mathbf{Z}$ (p. 184); exactness of the approximate initial probit (p. 189).
- *Implicit, in our reading:* τ independent of $\mathbf{Z}$; independence across individuals; the same stable process before and after J; no forward-looking behavior.

A.5.2 Keane and Wolpin (1997) and Card and Hyslop (2005): latent types with initial conditions

We work from the published article for Keane and Wolpin (journal pages) and from NBER Working Paper 10647 for Card and Hyslop, not the published Econometrica article, so Card–Hyslop page numbers are working-paper pages, tables are cited by number, and the published equations and estimates may differ. *Why it is included:* the two papers handle the state in which persistent types are first observed in two different ways, type shares conditional on the initial state and a sample in which everyone shares the same initial state, and a household first observed partway through owning a vehicle needs one of them.

(a) Keane and Wolpin: type shares conditional on initial schooling.

Setup. 1,373 white males, NLSY79 core sample, annual from age 16 through September 1988 (p. 483); five exclusive choices: white-collar, blue-collar, military, school, home (p. 477); terminal age $A = 65$ (p. 490). Finite-horizon structural model, simulated maximum likelihood with an approximated EMAX (p. 481; App. A). *Given state:* $\mathbf{S}(a) = \{\mathbf{e}(16), g(a), \mathbf{x}(a), \boldsymbol{\epsilon}(a)\}$, with schooling g and experience $\mathbf{x}$ observed by both, the endowment $\mathbf{e}(16)$ and shocks only by the agent. Wages are observed in chosen occupations.

Equations and derivation.

1. **Rewards** (3), p. 479: $w_m(a) = r_m \exp[e_m(16) + e_{m1}g(a) + e_{m2}x_m(a) - e_{m3}x_m^2(a) + \epsilon_m(a)]$, $m = 1, 2, 3$; $R_4(a) = e_4(16) - tc_1 I[g(a) \geq 12] - tc_2 I[g(a) \geq 16] + \epsilon_4(a)$; $R_5(a) = e_5(16) + \epsilon_5(a)$. The extended model adds mobility, re-entry, diploma and age terms ((C1)–(C3), pp. 520–521). *Assumptions used:* $\boldsymbol{\epsilon}(a) \sim N(0, \Omega)$, serially uncorrelated (p. 479, fn. 9).

2. **Dynamic program** (5)–(6), p. 480: $V_m(\mathbf{S}(a), a) = R_m(\mathbf{S}(a), a) + \delta E[V(\mathbf{S}(a+1), a+1) \mid \mathbf{S}(a), d_m(a) = 1]$ for $a < A$, $V_m = R_m$ at A, $V = \max_m V_m$. *Assumptions used:* deterministic accumulation $x_m(a+1) = x_m(a) + d_m(a)$, $g(a+1) = g(a) + d_4(a)$ (p. 480).

3. **Likelihood without types** (7), p. 481, with $\bar{\mathbf{S}}(a)$ the state net of shocks and $c(a)$ the choice and, when working, the wage: $\mathbb{P}[c(16), \ldots, c(\bar{a}) \mid g(16), \mathbf{e}(16)] = \prod_{a=16}^{\bar{a}} \mathbb{P}[c(a) \mid \bar{\mathbf{S}}(a)]$. *Assumption used:* serial independence of shocks.

4. **Types** (pp. 481–482, 490): $K = 4$, each an endowment vector $\mathbf{e}_k(16) = \{e_{mk}(16) : m = 1, \ldots, 5\}$ of intercepts. If $g(16)$ were exogenous, the likelihood would be $\prod_n \sum_k \pi_k L_{nk}$ (p. 482). *Assumptions used:* K known; technology, δ and Ω common to types; no military endowment heterogeneity (fn. 26).

5. **Initial-condition device** (8), p. 482. KW "assume that initial schooling is exogenous conditional on the age 16 endowment vector" (p. 482), and the intended contribution is

$$\mathbb{P}[c_n(16), \ldots, c_n(\bar{a}) \mid g_n(16)] = \sum_{k=1}^{K} \pi_{k|g_n(16)} \prod_{a=16}^{\bar{a}} \mathbb{P}[c_n(a) \mid g_n(16), \text{type} = k].$$

   Note: as printed, $\pi_{k|g}$ sits inside the product, a typesetting slip given the surrounding text. $\pi_{k|g}$ are free in two cells, $g(16) \in \{7, 8, 9\}$ and $\{10, 11\}$ (fn. 26, p. 490); $x_m(16) = 0$ (p. 479). Footnote 13 (p. 482): if data began at $a_0 > 16$, the likelihood would condition on $g(a_0)$ and $x_m(a_0)$, with type probabilities "estimated separately for each state combination."

6. **Equivalence with Heckman's conditional likelihood** (our reconstruction; not stated by KW). Bayes' rule gives $\mathbb{P}[c_n \mid g_n(16)] = \sum_k \pi_{k|g} L_{nk}$ with $\pi_{k|g} \equiv \pi_k \mathbb{P}[g \mid k] / \sum_{k'} \pi_{k'} \mathbb{P}[g \mid k']$, the discrete-type form of Heckman's (4.6) (Appendix A.5.1). With $\mathbb{P}[g \mid k]$ unrestricted, any free $\{\pi_{k|g}\}$ is generated by some $(\pi_k, \mathbb{P}[g \mid k])$, so (8) is exact, not an approximation like Heckman's probit reduced form. *Assumptions used:* exogeneity of $g(16)$ given type and serial independence (to write $\mathbb{P}[c_n \mid g_n(16), k] = L_{nk}$); saturation in the initial state. With many initial-state cells, pooling or parameterizing $\pi_{k|s}$ reintroduces approximation.

7. **Identification.** No argument is given for types, shares, or $\pi_{k|g}$. Types are disciplined by parametric structure and by wages, whose intercepts are the endowments (eq. (2), p. 478; App. B).

Results to note. Table 9 (p. 502, extended model): $\pi_{k|g} = (.0491, .1987, .4066, .3456)$ for nine years or less, $(.2343, .2335, .3734, .1588)$ for ten or more. In the basic model (Table B1 panel C) type

rankings differ, so labels are not comparable across specifications. Between-type variance is 90% of the variance of expected lifetime utility (p. 508); family background explains under 10% (Table 13). Within-sample χ^2 (Table 5, p. 495; not adjusted for estimated parameters): basic model rejects strongly, extended fits at most ages. Out of sample (Table 10, p. 505), white-collar shares track the CPS (.388 vs .413 at 30–33); blue-collar is overstated (.547 vs .460). Not shown: any estimate without the device, a test of exogeneity given type, or sensitivity to K.

Assumptions: stated and implicit.

- *Stated by the authors:* $K = 4$ known; initial schooling exogenous given the endowment; serially independent normal shocks; types shift intercepts only; free $\pi_{k|g}$ in two cells.
- *Implicit, in our reading:* any persistent unobservable that drove $g(16)$ other than the type is ruled out; a counterfactual that changes the distribution of $g(16)$ is outside the model; wage data supply identifying information that a choice-only panel lacks.

(b) Card and Hyslop: a common initial state by sample design.

Setup. Self-Sufficiency Project experiment, British Columbia and New Brunswick (p. 4): single-parent Income Assistance (IA) recipients on welfare in at least 11 of the previous 12 months, randomized; monthly IA records for 69 months (p. 6). Program members could receive a three-year earnings subsidy only if they began full-time work within 12 months (p. 4). A search model (pp. 16–19) motivates but is not estimated. *Outcome:* $y_{it} = 1$ if on IA. *Unobserved:* α_i.

Equations and derivation.

1. **Sample restriction** (p. 6). The 5,617 people on IA in *both* months before random assignment. "This restriction eliminates any 'initial conditions' problems associated with the second order dynamic models" (fn. 10); with $y_{i0} = y_{i,-1} = 1$ for all, the distribution of initial conditions need not be modeled (fn. 28, p. 21).

2. **Dynamic logit with random effect** (1), p. 21, with x_{it} a fourth-order polynomial in months since assignment:

$$P(y_{i1}, \ldots, y_{iT} \mid x_{i1}, \ldots, x_{iT}) = \int \Big\{ \prod_t L(\alpha_i + x_{it}\beta + \gamma_1 y_{it-1} + \gamma_2 y_{it-2} + \gamma_3 y_{it-1} y_{it-2}) \Big\} f(\alpha_i)\, d\alpha_i.$$

   f is $N(0, \sigma_\alpha^2)$ (10-point quadrature, fn. 29) or discrete with estimated mass points (p. 21). Preferred control specifications let the state-dependence terms vary linearly with α_i (Table 4, cols. 3–4). *Assumptions used:* second-order Markov given α_i; no baseline covariates (fn. 31).

3. **Why the restriction works** (our reconstruction). Heckman's conditional likelihood with $s_{i0} = (y_{i0}, y_{i,-1})$ is $\int \prod_t L(\cdot) f(\alpha \mid s_{i0})\, d\alpha$. If $s_{i0} = (1, 1)$ for everyone, f in (1) *is* $f(\alpha \mid s_0 = (1, 1))$: the problem is absorbed into the definition of the population, the stock of long-term recipients (fn. 8). *Assumptions used:* the conditioning set equals the full Markov state; a third-order model or baseline covariates would require $f(\alpha \mid s_0, x_{i0})$.

4. **Randomization** (pp. 16, 27): the "crucial" maintained assumption is that f is the same in both arms; it, not an initial-state model, identifies treatment effects. The restriction excluded 40 program and 27 other members, arm difference $p = 0.10$ (fn. 10).

5. **Program group** (2)–(5), pp. 27–28. IA and eligibility E_{it} load on the same α_i; (3) assumes eligibility independent of IA history given α_i. Eligibility hazard $\Phi[d(t) - k(\alpha_i)]$ while not yet

eligible and $t \le T_e = 14$, with $k(\alpha) = k_0\alpha$ or one value per mass point. Treatment enters the logit index through

$$\begin{aligned}\tau(t, E_{it}, t_i^e, y_{it-1}) = {} & E_{it}1(t_i^e \le t \le t_i^e + J - 1)\{\psi_0 1(y_{it-1} = 0) + \psi_1 1(y_{it-1} = 1)\} \\ & + E_{it}1(t_i^e + J \le t \le t_i^e + 35)\{\lambda_0 1(y_{it-1} = 0) + \lambda_1 1(y_{it-1} = 1)\}, \qquad J = 3.\end{aligned}$$

Entitlement effects are "directly identified" through λ_0, λ_1; establishment effects are not (p. 29).

Results to note. With three or more mass points, one goes to the boundary (value 16 or more), "pure stayers" (p. 24); five or six add little over four. Mass-point locations and probabilities are not reported in Tables 4 or 6. $k_0 = 0.21, 0.21, 0.29$ (Table 6, cols. 2–4); median α is -0.98 among eligibles and 0.36 among ineligibles (p. 30); treating eligibility as exogenous inflates entitlement effects by about 30% (p. 30). Control fit (informal Pearson over 50 history cells, fn. 33): 752.6 (first order), 260.3 (second), 253.0 (interactions), 175.8 (four mass points) (Table 4). Over 99% of the variance of mean monthly IA participation is explained (p. 33); the never-leaver share is mispredicted (p. 32). No out-of-sample forecast.

Assumptions: stated and implicit.

- *Stated by the authors:* common initial state by design; second-order Markov given α_i; f normal or discrete and common across arms; eligibility independent of IA history given α_i.
- *Implicit, in our reading:* the estimated f describes the sampled stock only, not welfare entrants; the device needs a conditioning event fixed before any outcome is recorded; any type-dependent event used for conditioning must be modeled jointly with the type, as (2)–(5) do for eligibility.

Comparison of the two devices (our reconstruction). Both are Heckman's conditional likelihood with an unrestricted initial-state distribution; CH's is KW's with one initial-state cell, so $\pi_{k|s_0} = \pi_k$ for the sampled population. KW's validity rests on an untested behavioral assumption (exogeneity given type) and applies to initial states that vary across people; CH's rests on the sampling rule and the Markov order, with randomization supplying arm comparability. Neither paper gives an identification argument for the mixture.

Question for discussion. For households first seen at vehicle age a and liquidity ℓ, a KW-style $\pi_\tau(a, \ell)$ needs panel entry exogenous given type and state: does a household enter the bank data independently of its vehicle decisions, or for example when it takes an auto loan, which would condition on a type-dependent purchase?

B Probability Targets and Interpretation of the Framework

These expressions retain the review's notation and clarify the objects discussed in the main text. They are probability identities or organizing representations, rather than additional identification results.

With the origin-conditioned hazards and survival probabilities in equations (1)–(3), the first-transaction purchase probability is

$$\mathbb{P}\Big(\widetilde{T}_h \le t_0 + H, J_h \in \mathcal{P} \mid \mathcal{I}_{h,t_0}\Big) = \sum_{k\in\mathcal{P}} F^{\mathrm{tr}}_{hk}(t_0, H). \tag{11}$$

It counts additions and replacements only when the first transaction is one of those events. A subsequent purchase after disposal requires a process that permits the intermediate transition. Leasing remains outside the working purchase endpoint.

The earlier Need–Ability–Choice formulation used a generic mixture representation. Let B_{hr} indicate purchase in decision period r, $\mathcal{I}^-_{hr}$ denote information immediately before that decision, and N_{hr} and A_{hr} denote possibly latent adjustment pressure and financial feasibility. For their possible values ν and α,

$$\mathbb{P}(B_{hr} = 1 \mid \mathcal{I}^-_{hr}) = \int \mathbb{P}(B_{hr} = 1 \mid \nu, \alpha, \mathcal{I}^-_{hr})\, dF_{N,A|\mathcal{I}^-_{hr}}(\nu, \alpha). \tag{12}$$

The joint distribution permits dependence between the two constructs. It supplies neither a measurement model nor an argument separating their effects from preferences or the value of waiting.

If instead N, A, and C are defined as binary conditions, the probability chain rule is

$$\mathbb{P}(N, A, C \mid X) = \mathbb{P}(N \mid X)\mathbb{P}(A \mid N, X)\mathbb{P}(C \mid N, A, X). \tag{13}$$

This identity does not assume independence. Equating purchase with the conjunction of these three conditions would be an additional behavioral restriction; requiring an objectively recorded need event, for example, would exclude discretionary purchases without one. The main text therefore uses the three terms to organize economic mechanisms rather than as mandatory independent gates.

C Reading Agenda and Coverage Limits

The following entries are outside the completed full-text review. The first three were previously listed among the selected papers but remain pending. Their proposed roles are provisional; no estimator or baseline is accepted on the basis of these descriptions.

C.1 de Lapparent and Cernicchiaro (2012)

Full-text review pending. Household-panel keep/replace and mileage model for French one-car households, 2000–2007, with income as a stochastic state (from the abstract). To verify: period length, shock distribution, treatment of β, and how income enters the state.

C.2 Arcidiacono and Miller (2011)

Full-text review pending. CCP estimation with discrete unobserved types, and possibly Markov unobserved states, via EM. To verify: whether it handles both time-invariant types and a Markov latent state with an initial-state distribution, and whether vehicle purchase satisfies its finite-dependence condition. Hotz and Miller (1993) use a *terminating* action (sterilization) to simplify the continuation value; whether a purchase that resets the fleet state plays the analogous role is the question this paper should settle. This determines whether conditional independence can be relaxed with a known estimator.

C.3 Shumway (2001)

Full-text review pending. Discrete-time hazard estimated as a multi-period logit, a candidate baseline. To verify: the equivalence result and the out-of-sample evaluation. Potential role: a reduced-form comparison, a horizon probability $1 - \prod_r \{1 - \lambda(r)\}$ built from period hazards $\lambda(r)$, whose event definition and evaluation would have to match the proposed purchase forecast.

C.4 Further references identified for later reading

A bounded search identified the references below. Their bibliographic entries were checked against publisher records or source pages, but these papers have not yet been read in full. The descriptions reflect abstracts and front matter and explain their possible relevance. They are a reading agenda, not additional findings supporting the main text. The C1–C5 labels correspond to the groups in Appendix A.

Selected for full review next

These five are priorities for full-text assessment; their findings and assumptions remain to be checked.

- Green et al. (2020): the Cash-for-Clunkers rebate as down-payment liquidity rather than credit capacity. It bears on how liquidity enters the purchase decision (C4).
- Baugh et al. (2021): even highly liquid households spend out of tax refunds. A test of whether an account balance is the right liquidity state (C4).
- Adda and Cooper (2000b): a household keep-or-replace model of car purchases with age and income shocks, aggregated to sales (C1).

- Keane and Wolpin (2007): validating a structural model on a holdout sample that faces a different policy regime (evaluation).
- Haider et al. (2020): calibration of predicted survival distributions at a fixed horizon (evaluation).

C1. Dynamic choice of when to buy: durable and vehicle replacement

- Attanasio et al. (2022): a life-cycle model separating whether to buy a car from how much to spend; purchases respond to expected income and its variability.
- Gillingham et al. (2018): a finite-horizon vehicle ownership, type, and usage model on Danish registers, with an explicit scrap-or-sell choice.
- Berger and Vavra (2015): a durable adjustment hazard estimated on household panel data; a reduced-form counterpart to the purchase probability.
- Bajari et al. (2013): a household-panel template for nonconvex transaction costs with durable, liquid-asset, and income states (housing).
- Gavazza et al. (2014), Chen et al. (2013), Stolyarov (2002): persistent differences in how much households value newer cars, lumpy transaction costs, and resale behavior that is not monotone in age. Together they bear on the latent-type interpretation and on the shape of the age effect.
- Copeland (2014): buyers shift purchase timing within the model year in response to prices; seasonality at a six-month horizon.
- Jacobsen and van Benthem (2015): the elasticity of scrappage with respect to used-car value.
- Glerum et al. (2013), Wakamori (2011), Archsmith et al. (2020): adding versus replacing a car in multi-vehicle households.
- Liu and Cirillo (2018), Bierlaire et al. (2021): dynamic choice models of vehicle purchase timing in transportation research.

C2 and C5. Identification, estimation, latent types, and initial conditions

- Hu and Shum (2012): identification with a serially correlated unobserved state; the route if fixed types are relaxed to a changing latent state.
- Kasahara and Shimotsu (2014): a data-based lower bound on the number of mixture components; a check on assuming two types.
- Bonhomme et al. (2022): discrete groups as an approximation to continuous heterogeneity.
- Heckman and Singer (1984): mass-point heterogeneity in duration models.
- Norets and Tang (2014): how much counterfactual choice probabilities depend on the extreme-value shock assumption.
- Arcidiacono and Miller (2019): finite dependence in nonstationary models.

- Aguirregabiria and Mira (2002), Aguirregabiria and Mira (2007), Su and Judd (2012), Iskhakov et al. (2016), Arcidiacono and Jones (2003): estimator choice (nested pseudo-likelihood, constrained optimization, nested fixed point, and EM with sequential likelihood).
- Chernozhukov et al. (2022), Adusumilli and Eckardt (2025): debiased and temporal-difference estimators built from first-stage choice probabilities.
- Arcidiacono and Ellickson (2011), Keane et al. (2011): surveys of estimation methods for dynamic discrete choice.

C4. Liquidity and purchase timing

- Parker et al. (2013), Mian and Sufi (2012), Hoekstra et al. (2017), Berger et al. (2025): fiscal payments and vehicle-purchase programs that move purchases forward in time, including later reversal.
- Di Maggio et al. (2017), Stephens (2008), Coulibaly and Li (2006): predictable cash-flow changes from mortgage resets and loan payoffs, dated in a lender's own records.
- Aaronson et al. (2012): spending after minimum-wage increases concentrated in vehicle purchases.
- Agarwal and Qian (2014): spending that responds before the cash arrives.
- Farrell and Greig (2019): tax-refund inflows and durable outflows observed in Chase account data.
- Olafsson and Pagel (2018), Ganong and Noel (2019): measuring liquidity and spending from account and transaction data.

Evaluation and predictive baselines

- Prinzie and Van den Poel (2007), Li et al. (2005), Kamakura et al. (2003): next-product-to-buy and cross-selling models in banking and retail; time since the last acquisition as a predictor.
- De Caigny et al. (2020), Mollaev et al. (2024), Babaev et al. (2022): predicting customer events from transaction sequences.
- Ishwaran et al. (2008), Lee et al. (2018), Kvamme and Borgan (2021): random survival forests and neural survival models.
- Gneiting and Raftery (2007), Gerds and Schumacher (2006): proper scoring rules, and the Brier score under censoring.
- Todd and Wolpin (2006), Iskhakov et al. (2020): when a structural model adds value over a predictive one.

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